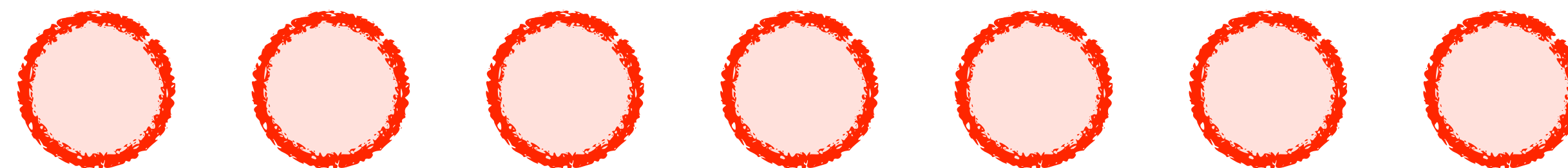


Advanced Algorithms

Hashing

刘明谋 Nanjing University, Suzhou, 2025

Balls into Bins



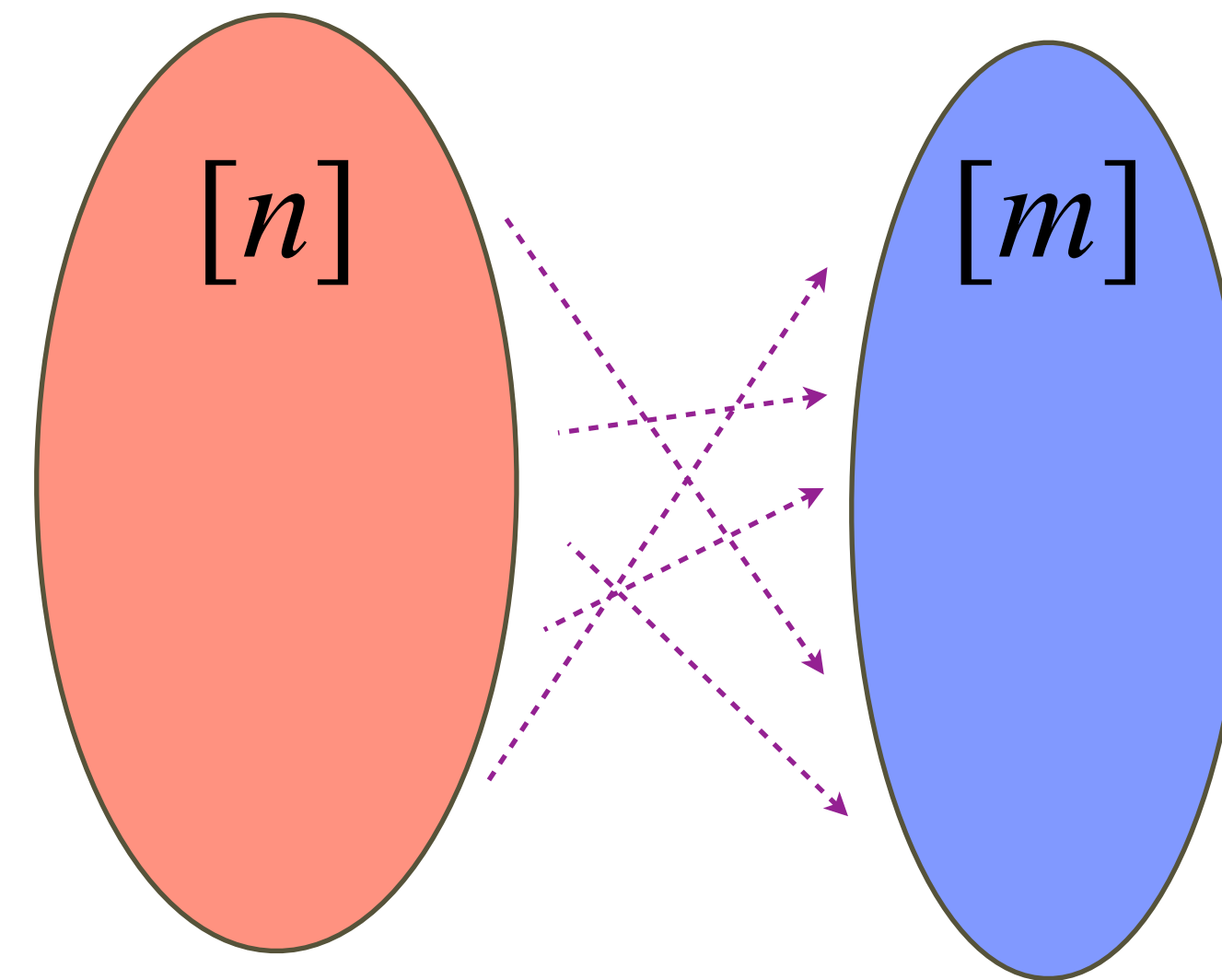
Balls into Bins (Random Function)

- n balls into m bins:

$$\Pr[\text{assignment}] = \frac{1}{m} \cdots \frac{1}{m} = \frac{1}{m^n}$$

- uniform random function:

$$\Pr[f] = \frac{1}{|[n] \rightarrow [m]|} = \frac{1}{m^n}$$

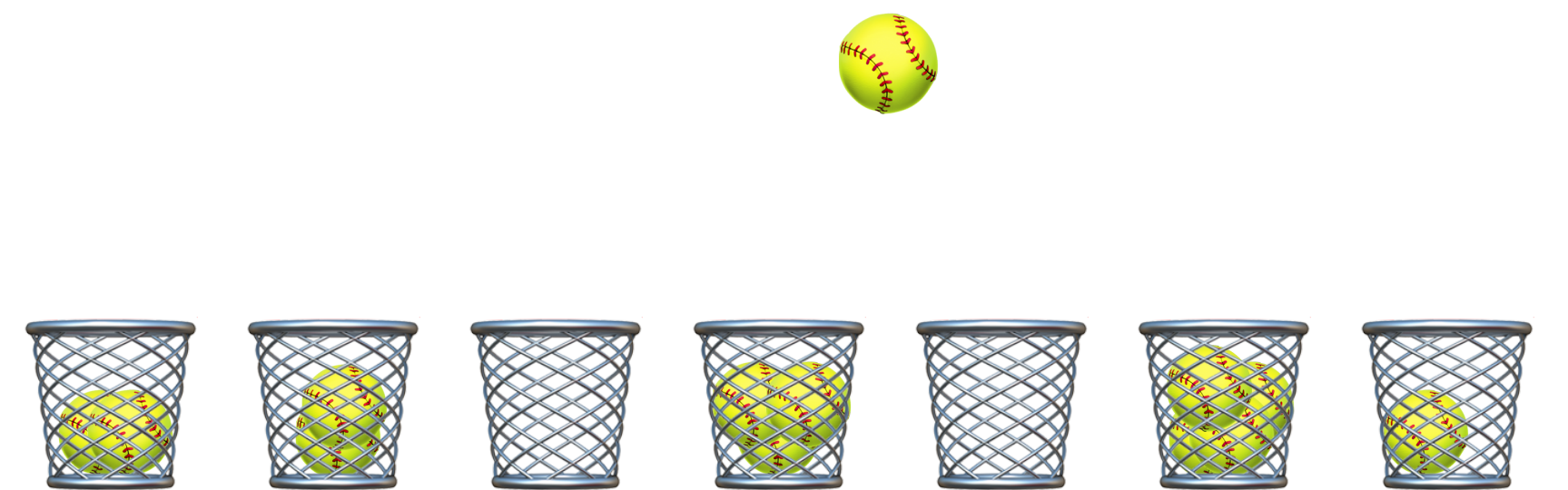


uniform random function

$$f : [n] \rightarrow [m]$$

1-1	birthday
on-to	coupon collector
pre-image size	occupancy

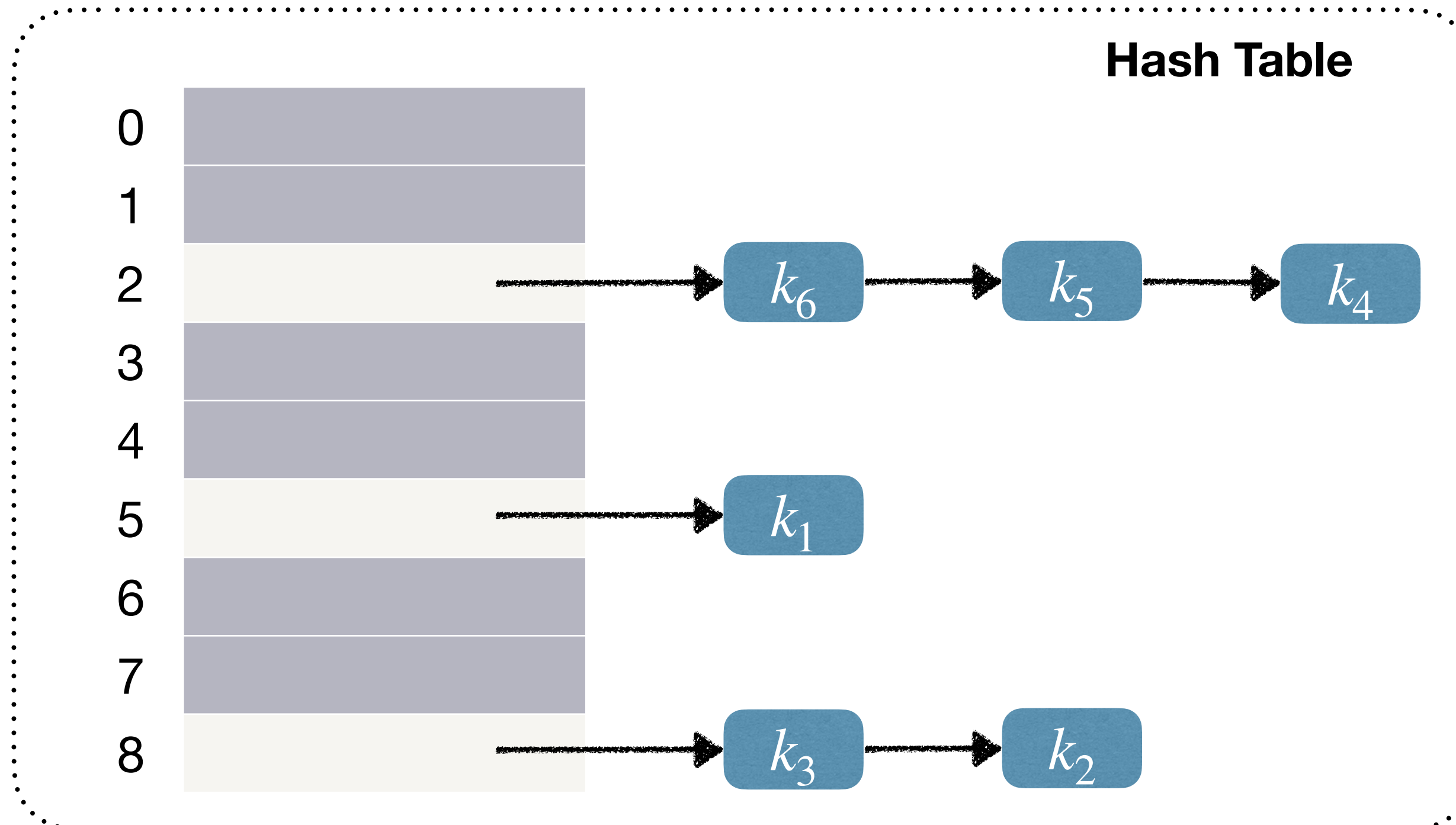
Load Balancing





Chaining

- Each bucket i stores a pointer to a linked list L_i .
- All keys that are hashed to index i go to L_i .



$$h(k_4) = h(k_5) = h(k_6) = 2$$

Space Cost:

- $\Theta(m)$ for pointers;
- $\Theta(n)$ for actual elements.



Hashing with Chaining

- **Search (k)** where k is a key. ----- Time depends on length of the linked list!
 - Compute $h(k)$
 - Go through the corresponding list to search item with key k .
- **Insert (x)** where x is a pointer to an item. ----- $O(1)$ Note: we assume computing h takes $O(1)$ time.
 - Compute $h(x.key)$
 - Insert x to the head of the corresponding list.
- **Remove (x)** where x is a pointer to an item. ----- $O(1)$
 - Simply remove x from the linked list.

Search can cost $\Theta(n)$ in worst-case. (All keys hash to same value.)

Balls into Bins

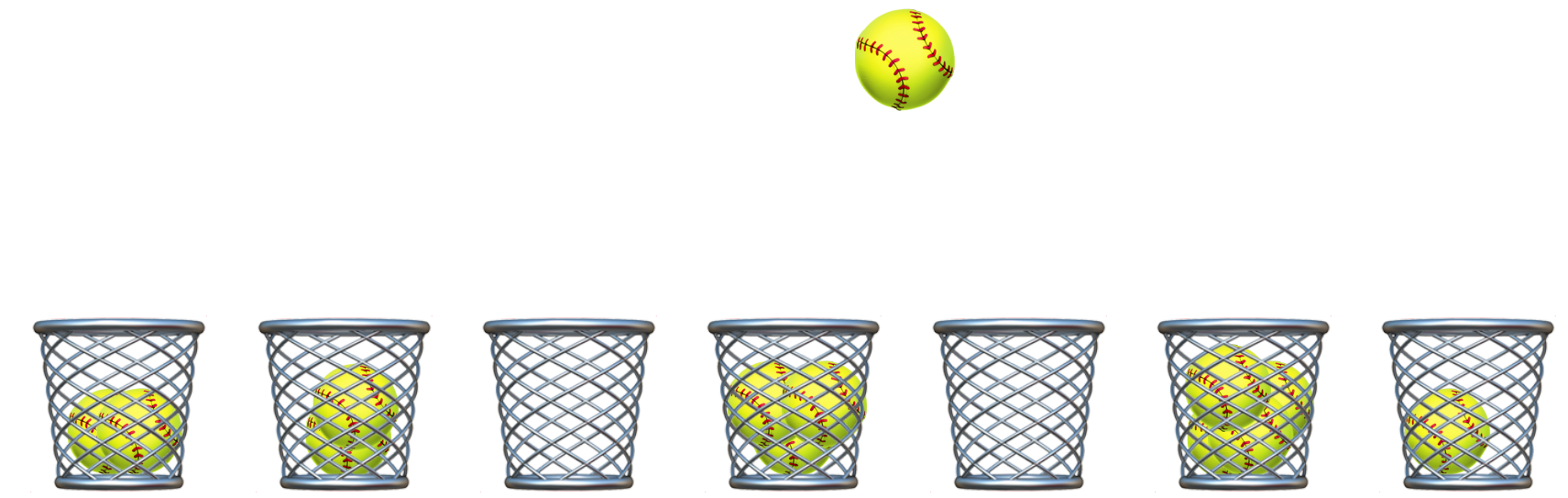
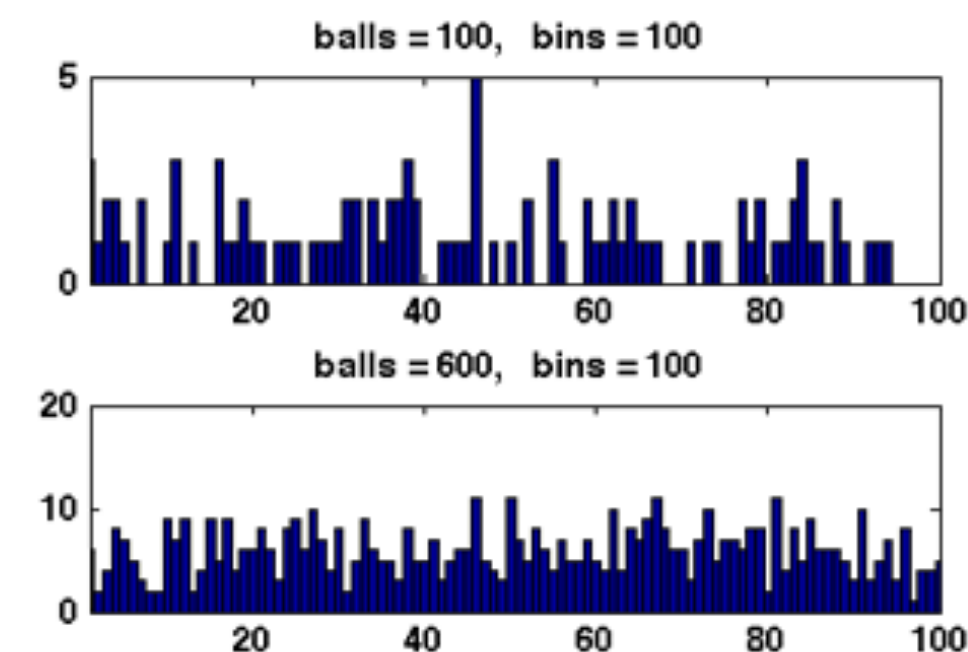
(Multinomial distribution)

- Throw m balls into n bins *u.a.r.* Numbers of balls received in each bins:

$$(X_1, X_2, \dots, X_n) \sim \text{multinomial distribution of parameters } m, \underbrace{(1/n, \dots, 1/n)}_n$$

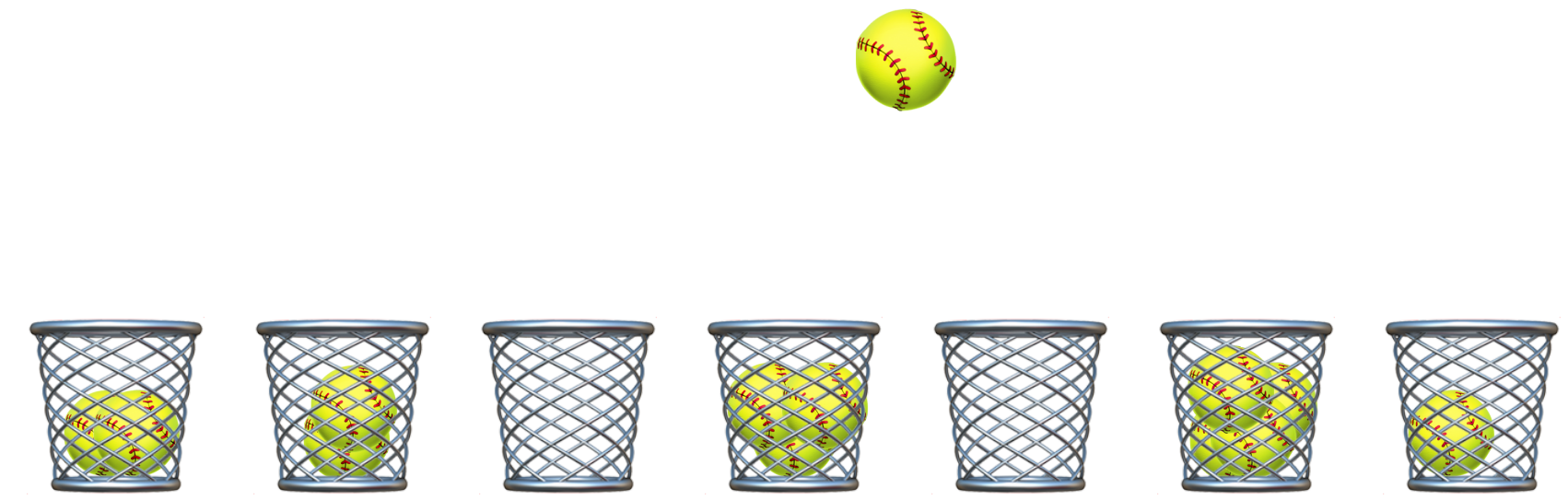
- Occupancy problem: the maximum load $\max_{1 \leq i \leq n} X_i$ *w.h.p.* (**with high probability**)

$$\max_{1 \leq i \leq n} X_i = \begin{cases} O\left(\frac{\log n}{\log \log n}\right) & \text{if } m = n \\ O\left(\frac{m}{n}\right) & \text{if } m \geq n \ln n \end{cases} \quad \text{w.h.p. (with prob. } 1 - o\left(\frac{1}{n}\right))$$



Balls into Bins

(Multinomial distribution)

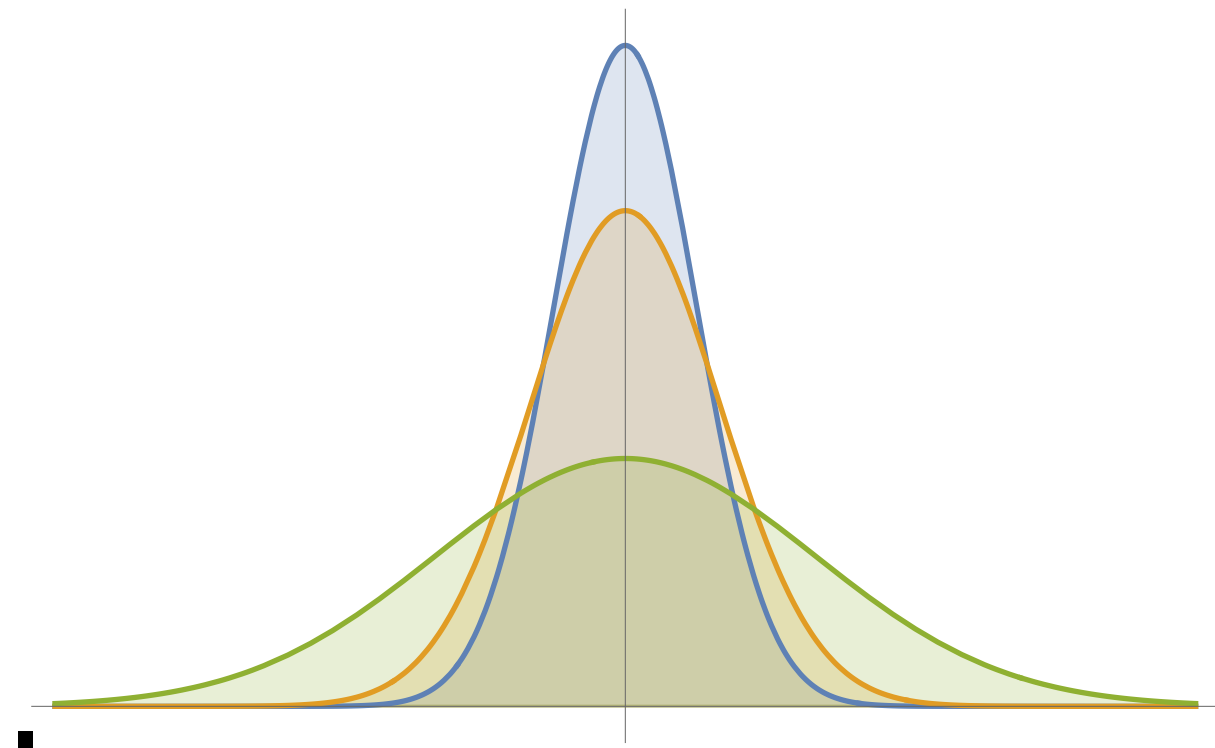


- Throw m balls into n bins *u.a.r.* Numbers of balls received in each bins:

$$(X_1, X_2, \dots, X_n) \sim \text{multinomial distribution of parameters } m, (1/n, \dots, 1/n)$$

- Occupancy problem: the maximum load $\max_{1 \leq i \leq n} X_i$ *w.h.p.* (**with high probability**)
- Markov: $\mathbb{E}[X_i] = 1$, $\Pr[X_i \geq k] \leq 1/k$. Union bound: $\Pr[\max_i X_i \geq k] \leq n/k$.
- Chebyshev: $X_i \sim \text{Bin}(n, 1/n)$. $\text{Var}(X_i) \leq 1$. $\Pr[|X_i - 1| \geq \sqrt{n}] \leq 1/n$.
- Binomial distribution: $X_i \sim \text{Bin}(n, 1/n)$, $\Pr[X_i \geq k] = \sum_{j \geq k} \binom{n}{j} \left(\frac{1}{n}\right)^j \leq \binom{n}{k} \left(\frac{1}{n}\right)^k$
 - **Fact:** $\binom{n}{k} \leq \left(\frac{ne}{k}\right)^k$ $\binom{n}{k} \left(\frac{1}{n}\right)^k \leq \left(\frac{e}{k}\right)^k \leq \frac{1}{n^{O(1)}}$ if $k = \Omega(\log n / \log \log n)$.

Chernoff Bound



- Chernoff bound: Let $X_1, \dots, X_n \in \{0,1\}$ be independent trials

$$X = \sum_{i=1}^n X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

(Poisson binomial RV
with mean μ)

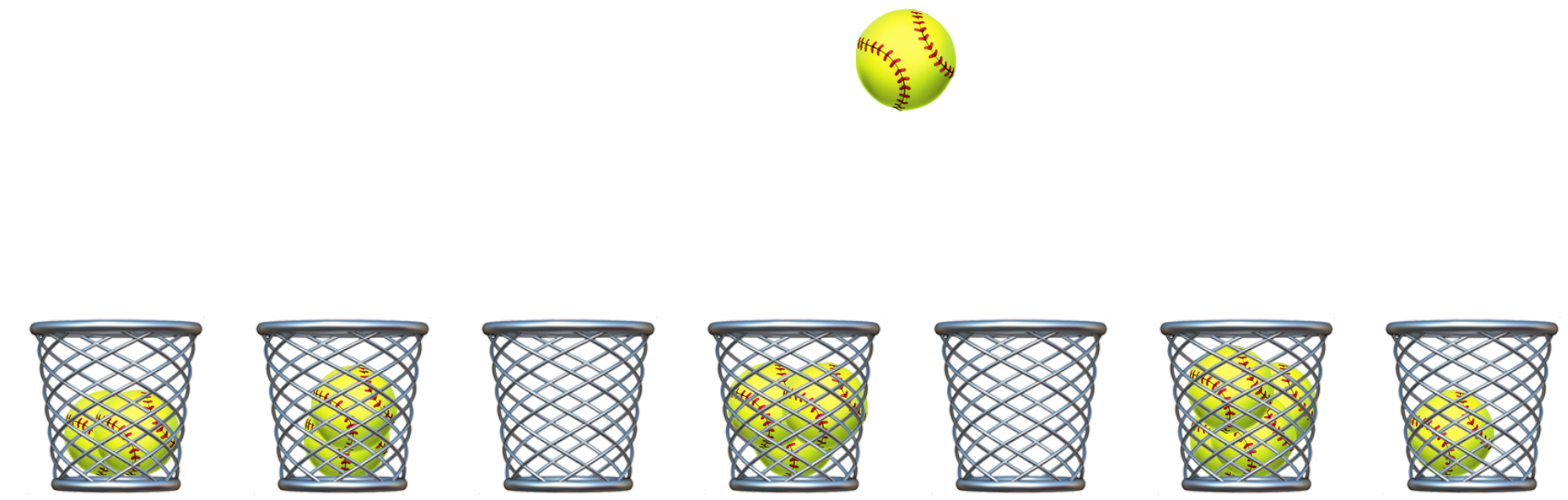
- For any $\delta > 0$,

$$\Pr(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$$

- For any $0 < \delta < 1$,

$$\Pr(X \leq (1 - \delta)\mu) \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu$$

Occupancy Problem



- Throw m balls into n bins *u.a.r.* The i -th bin receives X_i balls:

Marginally, $X_i \sim \text{Bin}(m, 1/n)$ and $\mu = \mathbb{E}[X_i] = \frac{m}{n}$

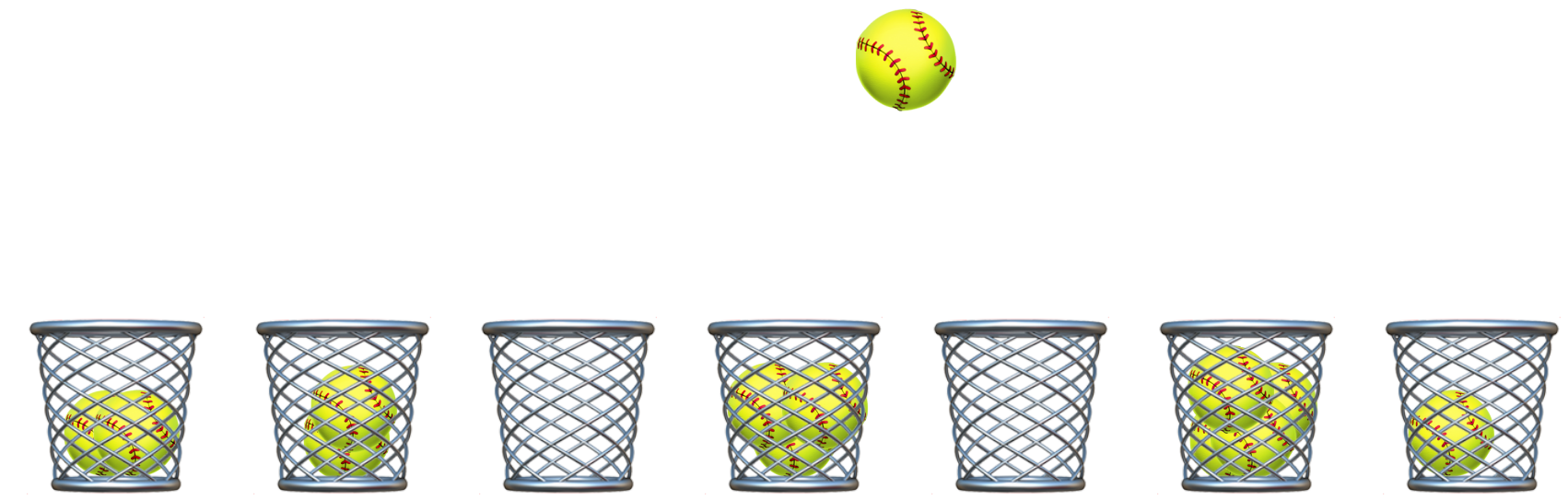
- When $m = n$: $\mu = 1$

$$\Pr(X_i \geq L) = \Pr(X_i \geq L\mu) \leq \frac{e^L}{eL^L} \leq \frac{1}{n^2} \quad \text{for } L = \frac{e \ln n}{\ln \ln n}$$

Chernoff: $\Pr(X_i \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1 + \delta)}} \right)^\mu$

- Union bound: $\Pr\left(\max_{1 \leq i \leq n} X_i \geq L\right) \leq \sum_{i=1}^n \Pr(X_i \geq L) \leq \frac{1}{n} \implies \max_{1 \leq i \leq n} X_i = O\left(\frac{\log n}{\log \log n}\right)$ **w.h.p.**

Occupancy Problem



- Throw m balls into n bins *u.a.r.* The i -th bin receives X_i balls:

Marginally, $X_i \sim \text{Bin}(m, 1/n)$ and $\mu = \mathbb{E}[X_i] = \frac{m}{n}$

- When $m \geq n \ln n$: $\mu \geq \ln n$

$$\Pr\left(X_i \geq \frac{2em}{n}\right) = \Pr(X_i \geq 2e\mu) \leq 2^{-2e\mu} \leq 2^{-2e \ln n} \leq \frac{1}{n^2}$$

Chernoff: $\Pr(X_i \geq L) \leq 2^{-L}$ if $L \geq 2e\mu$

- Union bound: $\Pr\left(\max_{1 \leq i \leq n} X_i \geq \frac{2em}{n}\right) \leq \sum_{i=1}^n \Pr\left(X_i \geq \frac{2em}{n}\right) \leq \frac{1}{n} \Rightarrow \max_{1 \leq i \leq n} X_i = O\left(\frac{m}{n}\right)$
w.h.p.

Chernoff Bound (Upper Tail)

- Let $X_1, \dots, X_n \in \{0,1\}$ be independent trials with $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$

$$\text{For any } \delta > 0: \quad \Pr(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$$

Proof: $\Pr(X \geq (1 + \delta)\mu) \leq \Pr(e^{tX} \geq e^{t(1+\delta)\mu}) \quad (\text{for any } t > 0)$

(Markov's inequality) $\leq e^{-t(1+\delta)\mu} \cdot \mathbb{E}[e^{tX}]$

Moment Generating Function

- The moment generating function (MGF) of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{k \geq 0} \frac{t^k \mathbb{E}[X^k]}{k!}$$

- If $X_1, \dots, X_n \in \{0,1\}$ are independent with $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$, then

$$M_X(t) = \mathbb{E}[e^{tX}] \leq e^{(e^t-1)\mu}$$

Proof: $\mathbb{E}[e^{tX}] = \mathbb{E}\left[\prod_{i=1}^n e^{tX_i}\right] = \prod_{i=1}^n \mathbb{E}[e^{tX_i}] = \prod_{i=1}^n (e^t p_i + (1 - p_i)) \leq \prod_{i=1}^n e^{(e^t-1)p_i}$

(independence) (where $p_i = \mathbb{E}[X_i]$) (because $1 + x \leq e^x$)

$= e^{\sum_{i=1}^n (e^t-1)p_i} = e^{(e^t-1)\mu}$ (where $\mu = \sum_{i=1}^n p_i$)

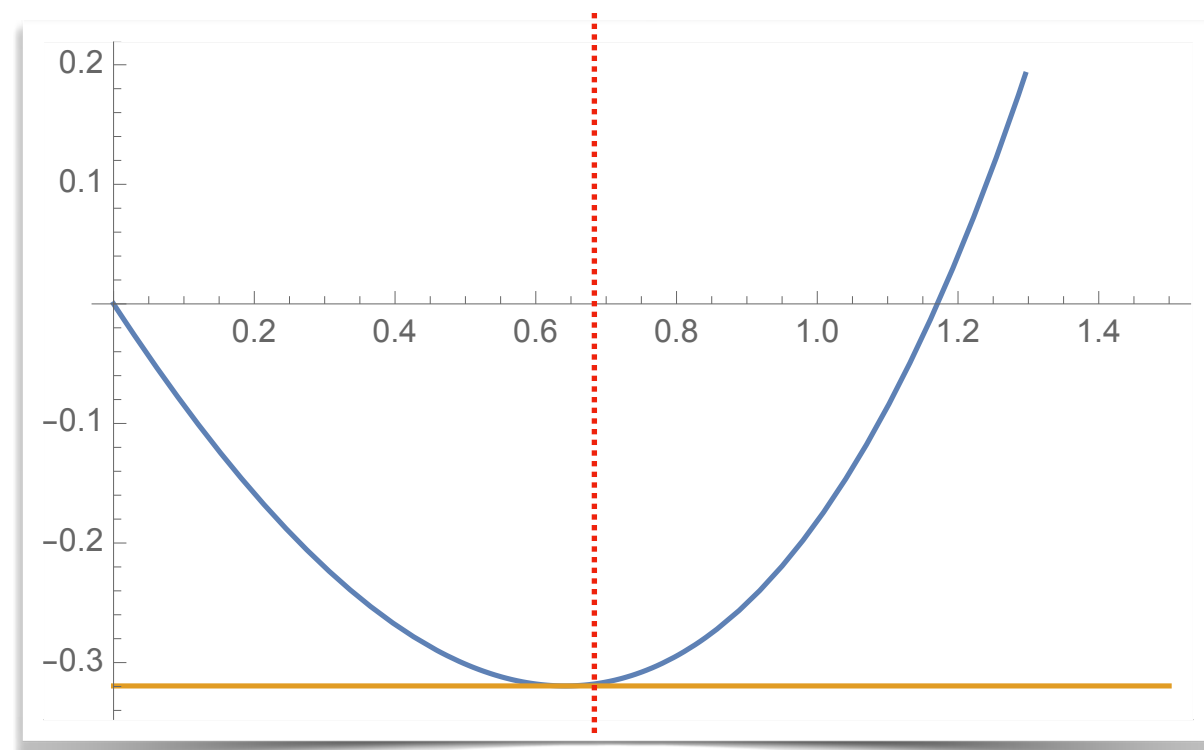
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- Let $X_1, \dots, X_n \in \{0,1\}$ be independent trials with $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$

$$\text{For any } \delta > 0: \quad \Pr(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$$

Proof: $\Pr(X \geq (1 + \delta)\mu) \leq \Pr(e^{tX} \geq e^{t(1+\delta)\mu}) \quad (\text{for any } t > 0)$

(Markov's inequality) $\leq e^{-t(1+\delta)\mu} \cdot \mathbb{E}[e^{tX}] \leq e^{-t(1+\delta)\mu} \cdot e^{(e^t-1)\mu}$



$$= e^{(e^t-1-t(1+\delta))\mu} = \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$$

(choose $t = \ln(1 + \delta)$)

(minimized at stationary point $t = \ln(1 + \delta)$)

Chernoff Bound (Lower Tail)

- Let $X_1, \dots, X_n \in \{0,1\}$ be independent trials with $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$

$$\text{For any } 0 < \delta < 1: \quad \Pr(X \leq (1 - \delta)\mu) \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu$$

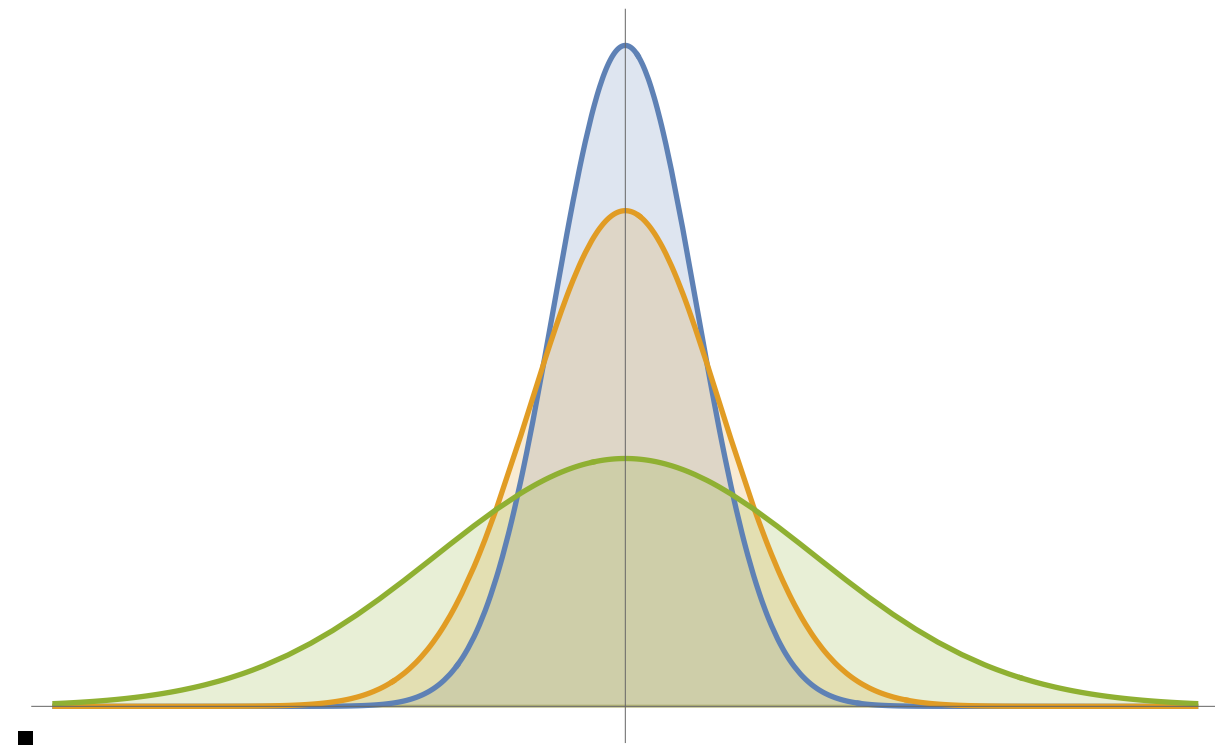
Proof: $\Pr(X \leq (1 - \delta)\mu) \leq \Pr(e^{tX} \geq e^{t(1-\delta)\mu})$ (for any $t < 0$)

(Markov's inequality) $\leq e^{-t(1-\delta)\mu} \cdot \mathbb{E}[e^{tX}] \leq e^{-t(1-\delta)\mu} \cdot e^{(e^t-1)\mu}$

$$= e^{(e^t-1-t(1-\delta))\mu} = \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu$$

(choose $t = \ln(1 - \delta)$)

Chernoff Bound



- Chernoff bound: Let $X_1, \dots, X_n \in \{0,1\}$ be independent trials

$$X = \sum_{i=1}^n X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

(Poisson binomial RV
with mean μ)

- For any $\delta > 0$,

$$\Pr(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu \leq \begin{cases} e^{-\frac{\mu\delta^2}{3}} & \text{if } 0 < \delta < 1 \\ 2^{-(1+\delta)\mu} & \text{if } (1 + \delta) \geq 2e \end{cases}$$

- For any $0 < \delta < 1$,

$$\Pr(X \leq (1 - \delta)\mu) \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu \leq e^{-\frac{\mu\delta^2}{2}}$$

Balls into Bins

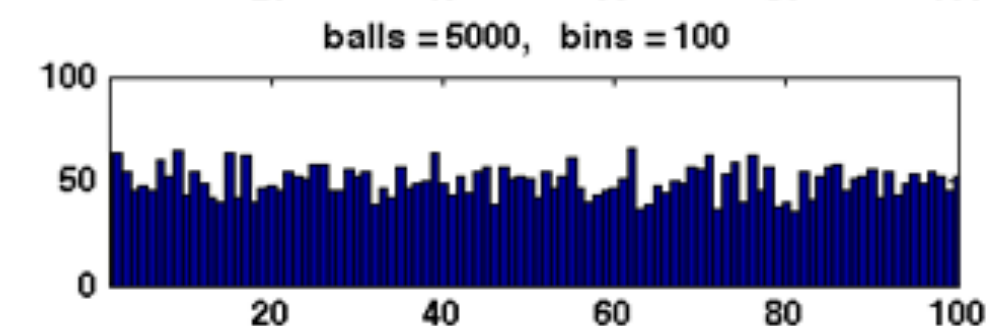
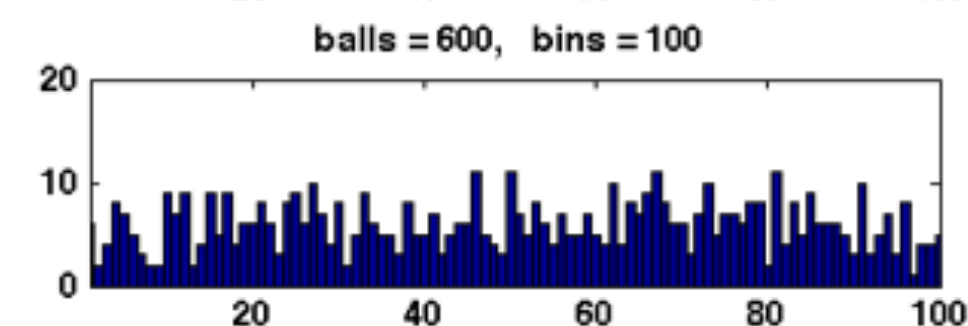
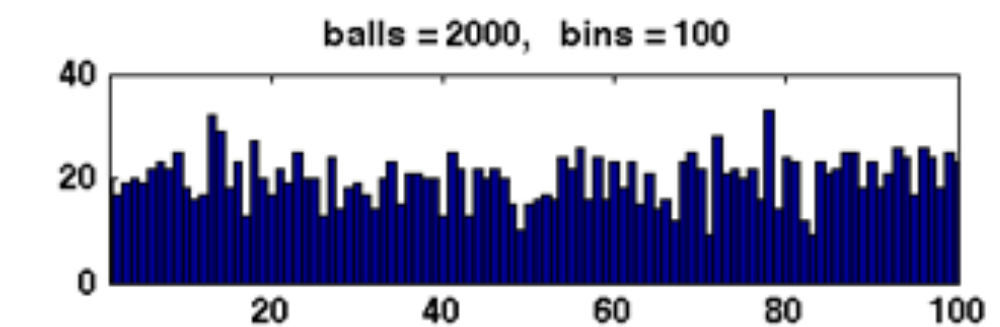
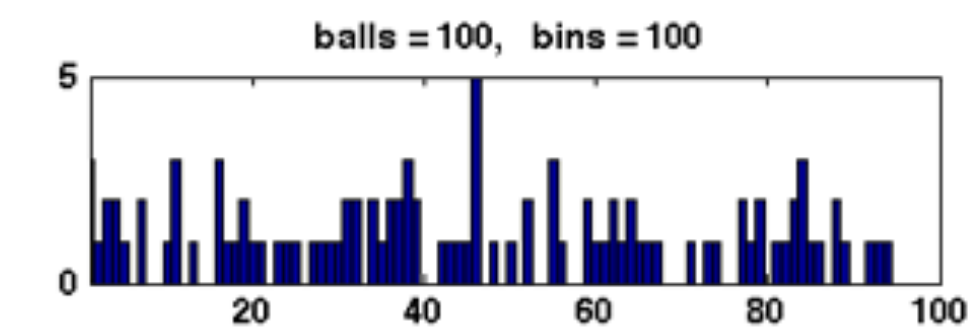
(Multinomial distribution)

- Throw m balls into n bins *u.a.r.* Numbers of balls received in each bins:

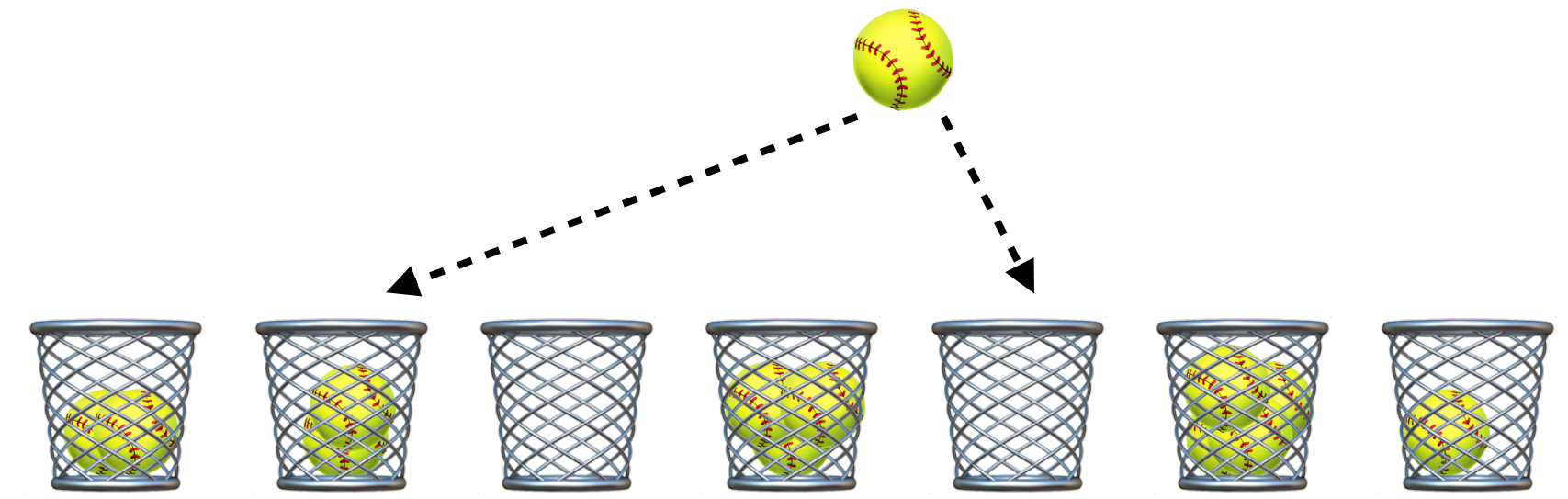
$$(X_1, X_2, \dots, X_n) \sim \text{multinomial distribution of parameters } m, \underbrace{(1/n, \dots, 1/n)}_n$$

- Occupancy problem: the maximum load $\max_{1 \leq i \leq n} X_i$ *w.h.p.* (**with high probability**)

$$\max_{1 \leq i \leq n} X_i = \begin{cases} O\left(\frac{\log n}{\log \log n}\right) & \text{if } m = n \\ O\left(\frac{m}{n}\right) & \text{if } m \geq n \ln n \end{cases} \quad \text{w.h.p. (with prob. } 1 - o\left(\frac{1}{n}\right))$$



Power of two choices



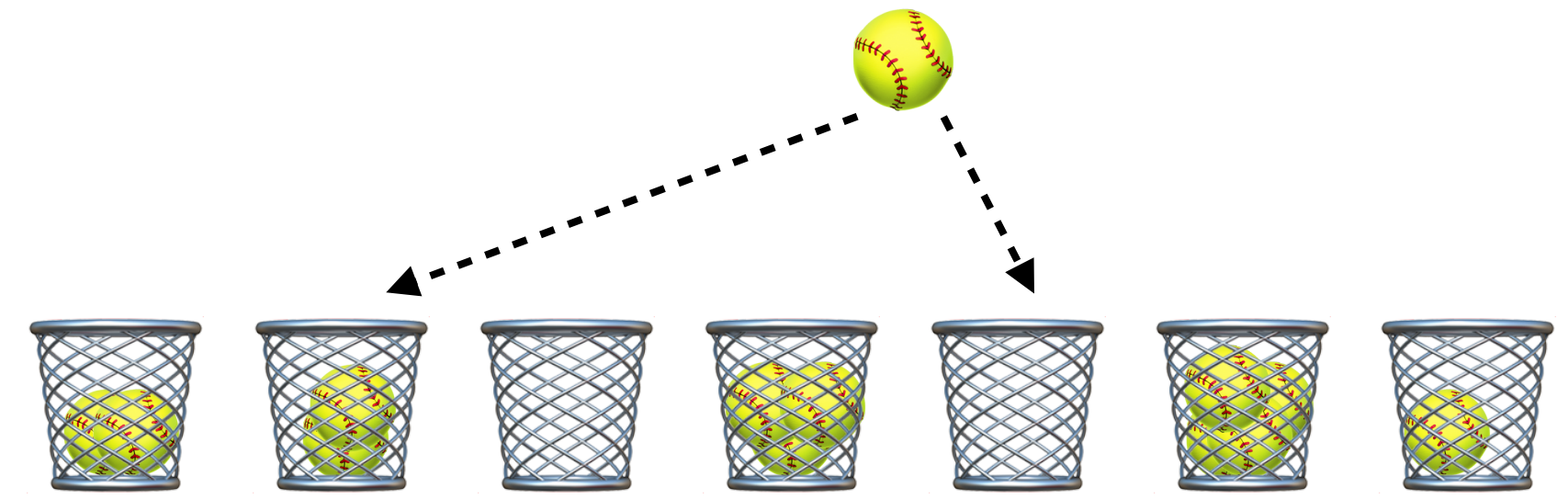
- Can we do better if we are equipped with **two** random functions?
- Idea: place the ball into the bin with less load.
- Intuitively better. Proof?

- Intuition: $B_i \triangleq \sum_x I(\text{Load}(x) \geq i)$

- $\mathbb{E}[B_i] \leq \sum_x \Pr[\text{Load}(x) \geq i] = n \cdot \Pr[\text{Load}(1) \geq i]$

$$\leq n \cdot (B_{i-1}/n)^2 = B_{i-1}^2/n$$

Power of two choices



- Can we do better if we are equipped with **two** random functions?
- Idea: place the ball into the bin with less load.
- Intuition: $B_i \triangleq \#\{\text{Load}(x) \geq i\} \approx n \cdot (B_{i-1}/n)^2 = B_{i-1}^2/n$
- $B_i/n \approx (B_{i-1}/n)^2$ and $B_2/n \leq 1/2$
- Intuitively: $\frac{B_{2+j}}{n} \leq \frac{1}{2^{2^j}}$. Max-load m satisfies $B_m/n < 1$.
- $2^{2^{\log \log n}} = n \implies m \approx \log \log n$

Power of two choices

Induction-like proof strategy: layer by layer

- Let $\beta_6 = n/2e$, $\beta_{i+1} = 2e\beta_i^2/n$ be thresholds, ν_i the #bins with load $\geq i$
- Let $\mathcal{E}_i$ be event $\nu_i \leq \beta_i$. **Goal:** $\Pr[\wedge_i \mathcal{E}_i] = \prod_i \Pr[\mathcal{E}_i | \mathcal{E}_{<i}] \geq 1 - O(1/n)$
- **Observation:** Given $\mathcal{E}_i$, $\nu_{i+1} \leq \#\{\text{balls with height} > i\} \lesssim \text{Bin}(n, \beta_i^2/n^2)$
- $\Pr[\text{Bin}(n, \beta_i^2/n^2) \geq \beta_{i+1}] \leq 2^{-2e\beta_i^2/n}$ **Chernoff:** $\Pr(X \geq L) \leq 2^{-L}$ if $L \geq 2e\mu$
 - $\beta_i^2/n \geq \log_2 n \implies \Pr[\neg \mathcal{E}_{i+1} | \mathcal{E}_i] \leq 1/n^{2e};$
 - o.w. $\nu_i < \sqrt{n \log_2 n}$, ν_{i+1} shrinks drastically: $\Pr[\nu_{i+1} \geq 6 \log n | \mathcal{E}_i] \leq 1/n^2$
 $\Pr[\nu_{i+2} \geq 1 | \nu_{i+1} < 6 \log n] \leq \Pr[\text{Bin}(n, 6 \log n/n)^2 \geq 1] = O(\log^2 n)/n$



Hashing with Chaining

- **Search (k)** where k is a key. - - - - - Time depends on length of the linked list!
 - Compute $h(k)$
 - Go through the corresponding list to search item with key k .
- **Insert (x)** where x is a pointer to an item. - - - - - $O(1)$ Note: we assume computing h takes $O(1)$ time.
 - Compute $h(x.key)$
 - Insert x to the head of the corresponding list.
- **Remove (x)** where x is a pointer to an item. - - - - - $O(1)$
 - Simply remove x from the linked list.

Search can cost $\Theta(n)$ in worst-case. (All keys hash to same value.)

Birthday Paradox

Paradox:

- (i) a statement that leads to a contradiction;
- (ii) a situation which defies intuition.



In a class of $m > 57$ students, with $>99\%$ probability, there are two students with the same birthday.

Assumption: birthdays are uniformly & independently distributed.

n balls are thrown into m bins:
event $\mathcal{E}$: each bin receives ≤ 1 balls

Birthday Paradox

n balls are thrown into m bins:

event $\mathcal{E}$: each bin receives ≤ 1 balls

$$\Pr[\mathcal{E}] = \frac{\left| [n] \xrightarrow{1-1} [m] \right|}{\left| [n] \rightarrow [m] \right|} = \frac{m(m-1)\cdots(m-n+1)}{m^n}$$

$$= \prod_{i=0}^{n-1} \left(1 - \frac{i}{m} \right)$$

Birthday Paradox

n balls are thrown into m bins:
event $\mathcal{E}$: each bin receives ≤ 1 balls

Suppose that balls are thrown one-by-one:

$$\Pr[\mathcal{E}] = \Pr[\text{all } n \text{ balls are thrown into distinct bins}]$$

chain rule $= \prod_{i=1}^n \Pr[\text{the } i\text{th ball is thrown into an empty bin} \mid$
first $i - 1$ balls are thrown into distinct bins]

$$= \prod_{i=1}^n \left(1 - \frac{i-1}{m}\right) = \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right)$$

Birthday Paradox

n balls are thrown into m bins:
event $\mathcal{E}$: each bin receives ≤ 1 balls

(Taylor: $1 - x \approx e^{-x}$ for $x = o(1)$)

$$\Pr[\mathcal{E}] = \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right) \approx \prod_{i=0}^{n-1} e^{-\frac{i}{m}} \approx e^{-n^2/2m}$$

Formally: $e^{-(1+o(1))n^2/2m} \leq \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right) \leq e^{-(1-o(1))n^2/2m}$
(assuming $n \ll m$)

when $n = \sqrt{2m \ln \frac{1}{p}}$ $\implies \Pr[\mathcal{E}] = (1 \pm o(1))p$

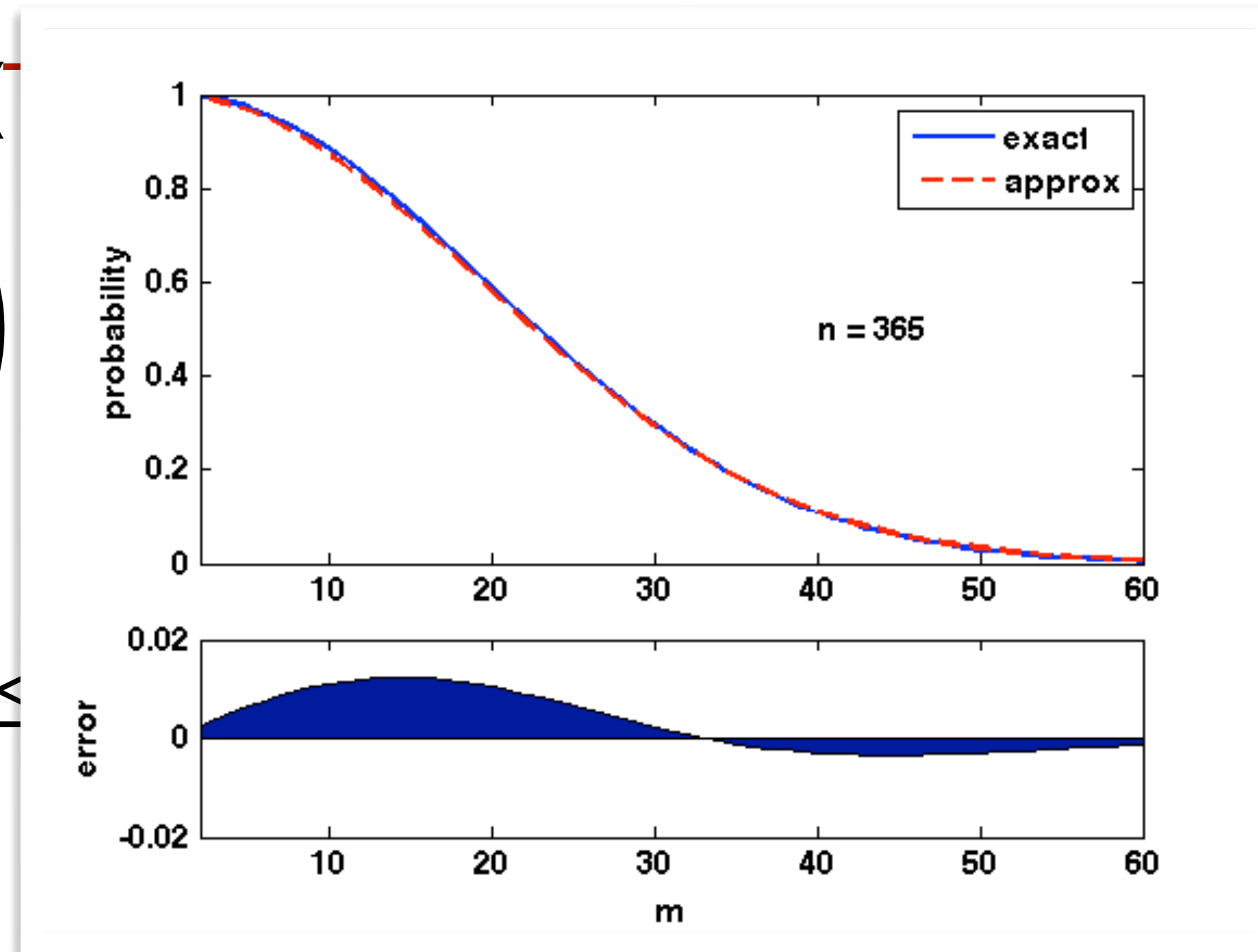
Birthday Paradox

n balls are thrown into m bins:
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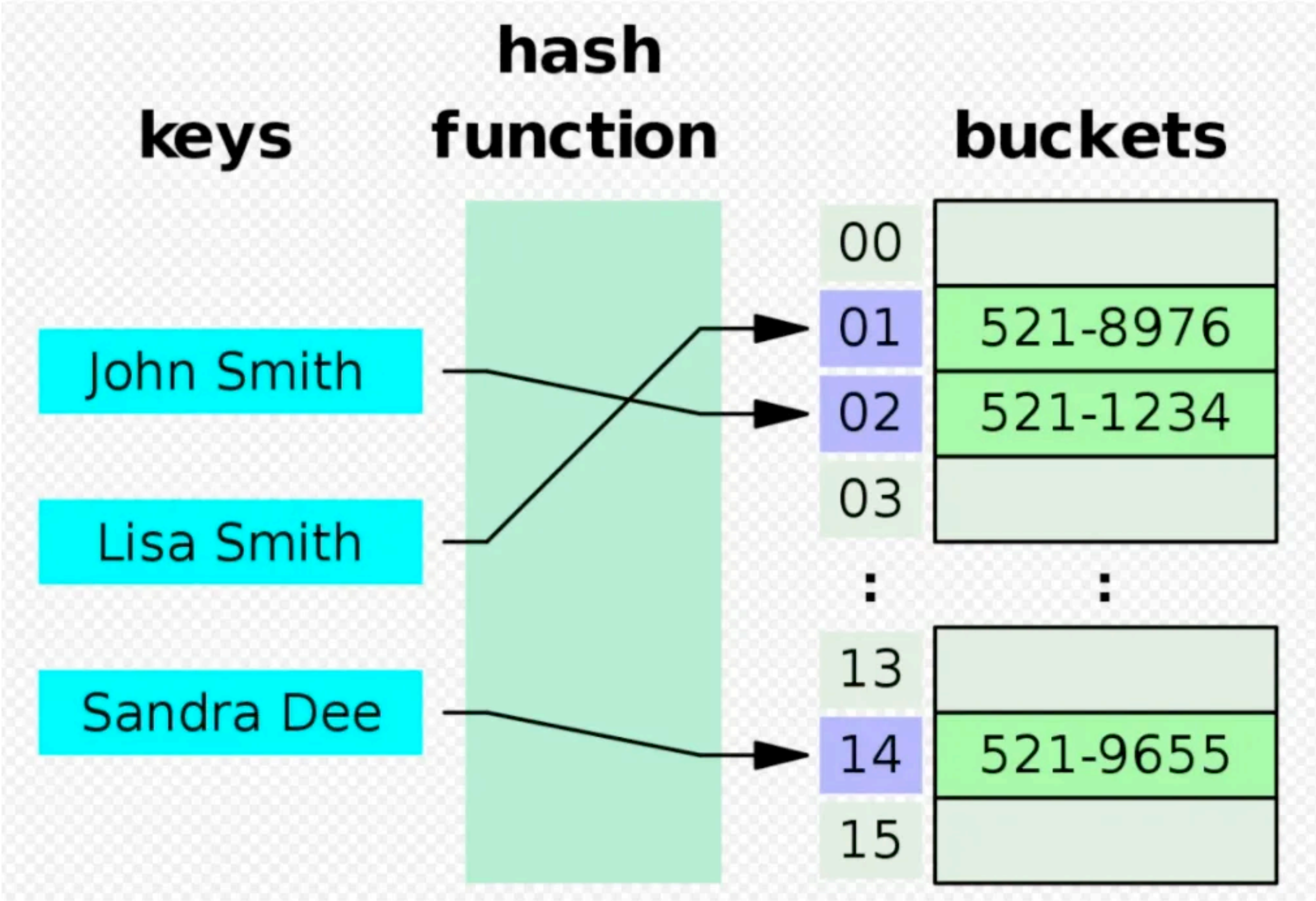
$$\Pr[\mathcal{E}] = \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right)$$

Formally: $e^{-(1+o(1))n^2/2m} \leq \Pr[\mathcal{E}] \leq e^{-(1-o(1))n^2/2m}$
(assuming $n \ll m$)

when $n = \sqrt{2m \ln \frac{1}{p}}$ $\implies \Pr[\mathcal{E}] = (1 \pm o(1))p$



Hash Table



Data Structure for Set

Data: a set S of n items $x_1, x_2, \dots, x_n \in U = [N]$

Query: an item $x \in U$

Determine whether $x \in S$.

- **Space cost:** size of data structure (in bits)
 - **entropy** of a set: $\log \binom{N}{n} = O(n \log N)$ bits (when $N \gg n$)
- **Time cost:** time to answer a query (in memory accesses)
- **Balanced tree:** $O(n \log N)$ bits of space, $O(\log n)$ time
- **Perfect hashing:** $O(n \log N)$ bits of space, $O(1)$ time

Perfect Hashing

$S = \{a, b, c, d, e, f\} \subseteq [N]$ of size n

uniform
random

h $[N] \rightarrow [m]$

no collision

$\Pr[\text{perfect}] \approx e^{-n^2/2m} > 1/2$

Table T :

e	b		d		f		c	a	
-----	-----	--	-----	--	-----	--	-----	-----	--

$m = n^2$

Birthday

SUHA: Simple Uniform Hash Assumption

Query(x):

retrieve hash function h ;

check whether $T[h(x)] = x$;

Universal Hashing

Universal Hash Family (Carter and Wegman 1979):

A family $\mathcal{H}$ of hash functions in $U \rightarrow [m]$ is **k -universal** if for any $\leq k$ distinct $x_1, \dots, x_k \in U$,

$$\Pr_{h \in \mathcal{H}} [h(x_1) = \dots = h(x_k)] \leq \frac{1}{m^{k-1}}.$$

Moreover, $\mathcal{H}$ is **strongly k -universal** (k -wise independent) if for any $\leq k$ distinct $x_1, \dots, x_k \in U$ and any $y_1, \dots, y_k \in [m]$,

$$\Pr_{h \in \mathcal{H}} \left[\bigwedge_{i=1}^k h(x_i) = y_i \right] = \frac{1}{m^k}.$$

k -Universal Hash Family

hash functions $h : U \rightarrow [m]$

- **Linear congruential hashing:**
 - Represent $U \subseteq \mathbb{Z}_p$ for sufficiently large prime p
 - $h_{a,b}(x) = ((ax + b) \bmod p) \bmod m$
 - $\mathcal{H} = \left\{ h_{a,b} \mid a \in \mathbb{Z}_p \setminus \{0\}, b \in \mathbb{Z}_p \right\}$

Theorem:

The linear congruential family $\mathcal{H}$ is 2-wise independent.

- **Degree- k polynomial in finite field with random coefficients**
- Hashing between binary fields: $GF(2^w) \rightarrow GF(2^l)$

$$h_{a,b}(x) = (a * x + b) \gg (w-1)$$

Birthday Paradox (pairwise independence)

n balls are thrown into m bins: **by 2-universal hashing**
event $\mathcal{E}$: each bin receives ≤ 1 balls

- Location of n balls: $X_1, X_2, \dots, X_n \in [m]$
- Total # of collisions:

$$Y = \sum_{i < j} I[X_i = X_j]$$

- Linearity of expectation:

$$\mathbb{E}[Y] = \sum_{i < j} \Pr[X_i = X_j] \leq \binom{n}{2} \frac{1}{m}$$

2-universal

Birthday Paradox (pairwise independence)

n balls are thrown into m bins: by 2-universal hashing
event $\mathcal{E}$: each bin receives ≤ 1 balls

- Location of n balls: $X_1, X_2, \dots, X_n \in [m]$
- Total # of collisions:

$$Y = \sum_{i < j} I[X_i = X_j]$$

- Linearity of expectation:

$$\mathbb{E}[Y] = \sum_{i < j} \Pr[X_i = X_j] \leq \binom{n}{2} \frac{1}{m}$$

2-universal

when

$$n \leq \sqrt{2m\epsilon}$$

- Markov's inequality: $\Pr[\neg \mathcal{E}] = \Pr[Y \geq 1] \leq \mathbb{E}[Y] \leq \epsilon$

Perfect Hashing

$S = \{a, b, c, d, e, f\} \subseteq [N]$ of size n

2-universal $\boxed{h} : [N] \rightarrow [m] \quad \Pr[\textit{imperfect}] = \frac{n(n-1)}{2m}$

Table T :

e	b		d		f		c	a	
-----	-----	--	-----	--	-----	--	-----	-----	--

 m

For 2-universal family $\mathcal{H}$ from $[N]$ to $[m]$, if $m > \binom{n}{2}$, for any $S \subseteq [N]$ of size n , there is an $h \in \mathcal{H}$ that cause no collisions over S .

Query(x):

retrieve hash function h ;

check whether $T[h(x)] = x$;

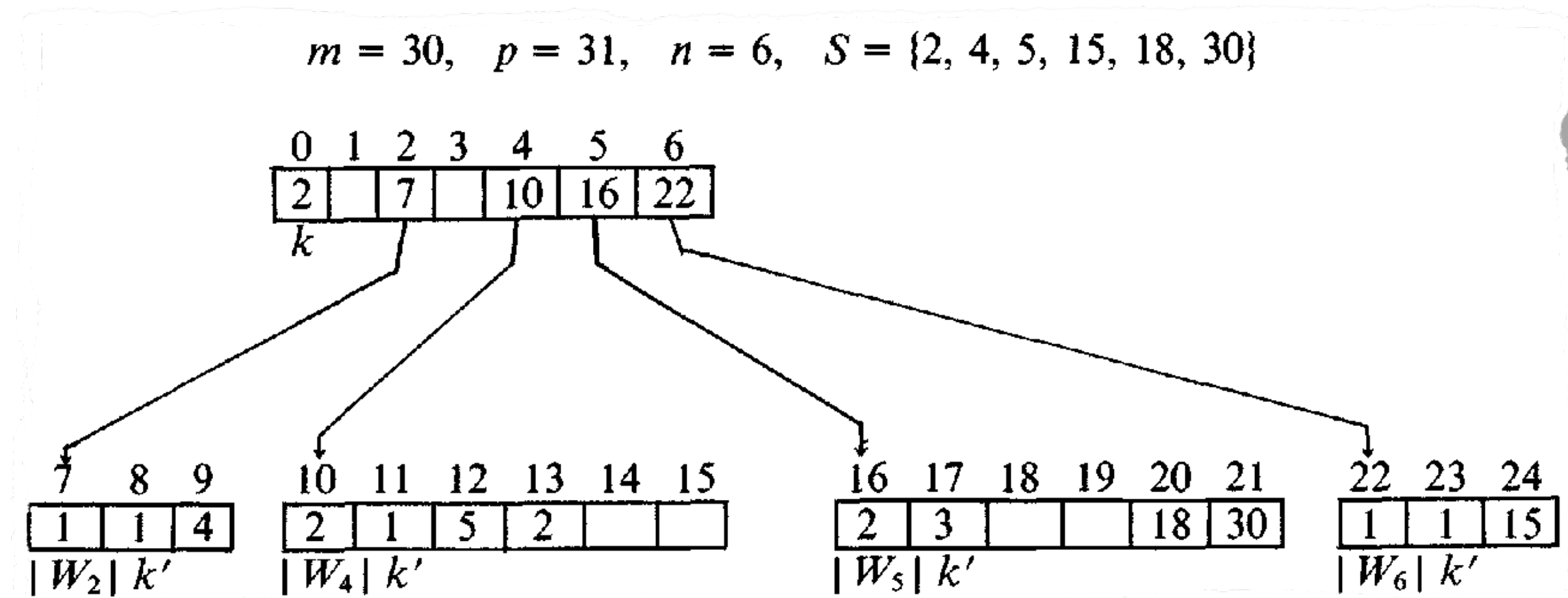
FKS Perfect Hashing

(Fredman, Komlós, Szemerédi, 1984)

Data: a set S of n items $x_1, x_2, \dots, x_n \in U = [N]$

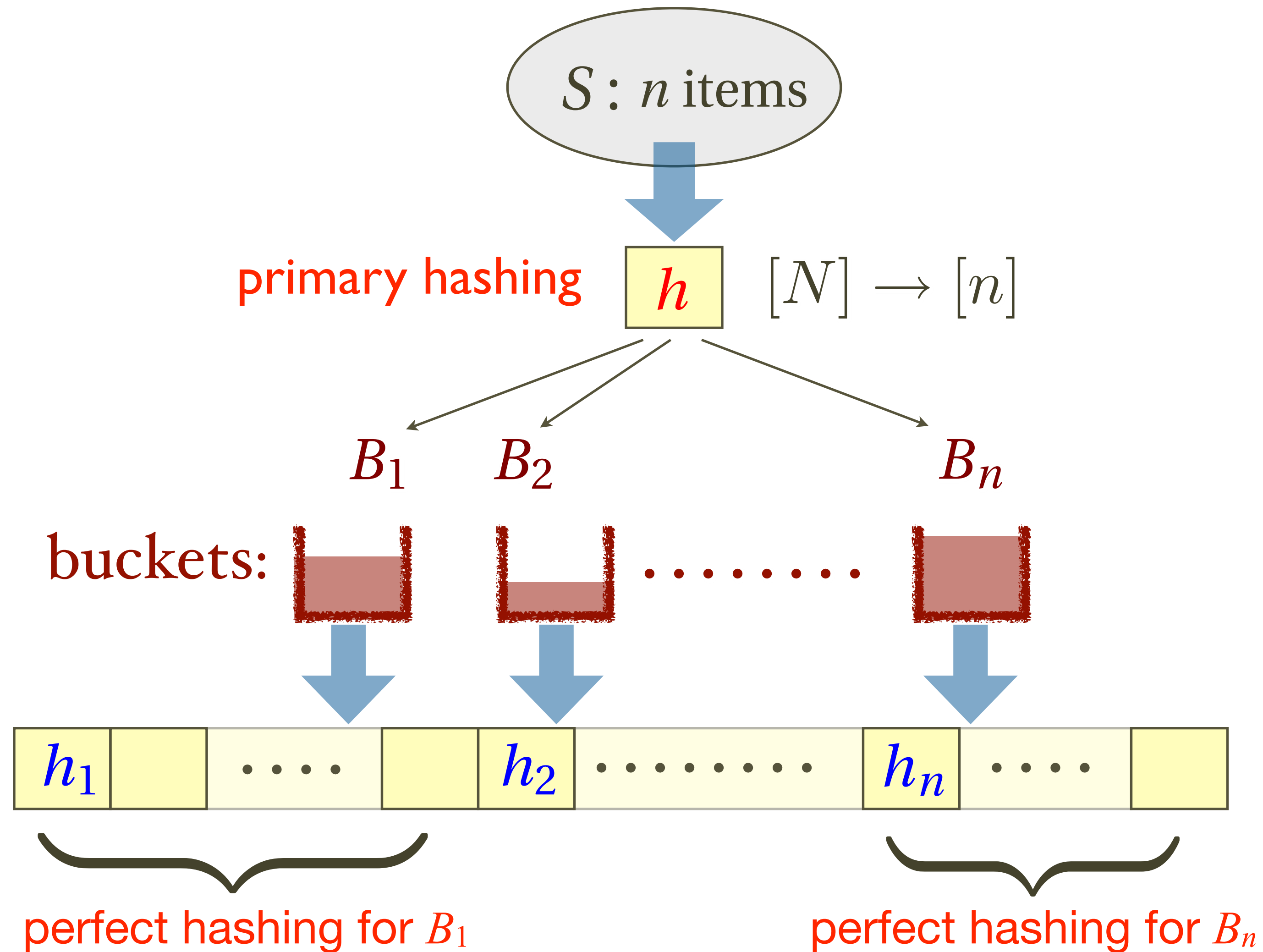
Query: an item $x \in U$

Determine whether $x \in S$.



- Space cost: $O(n)$ words (each of $O(\log N)$ bits)
- Time cost: $O(1)$ for each query in the worst case

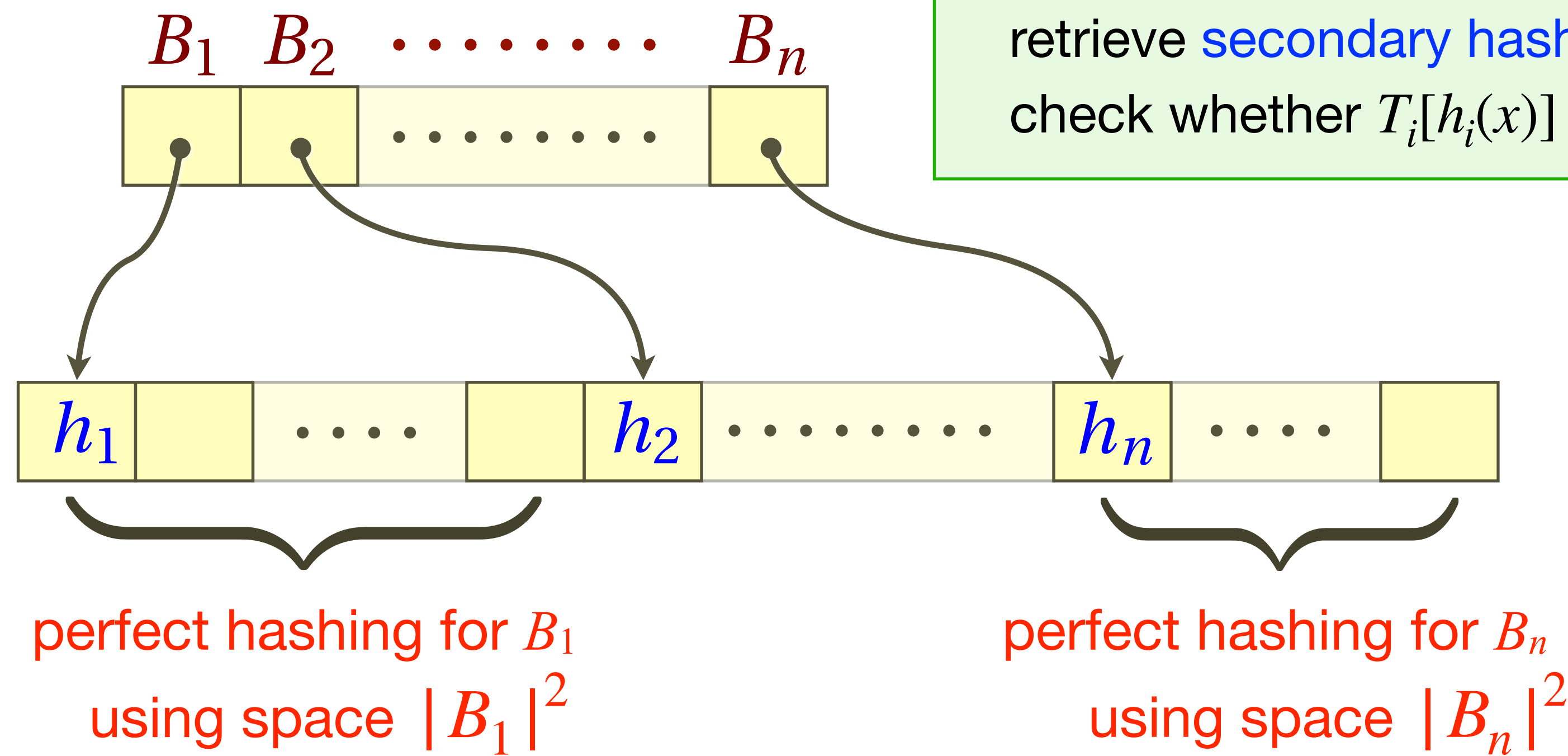
FKS Perfect Hashing



FKS Perfect Hashing

Set $S \subseteq [N]$ of size n

$$h: [N] \rightarrow [n]$$



Query(x):

retrieve **primary hash** h ;
goto **bucket** $i = h(x)$;
retrieve **secondary hash** h_i ;
check whether $T_i[h_i(x)] = x$;

- $\exists h_1, \dots, h_n$ from 2-universal family s.t. h_i is perfect for B_i for all i

Collision Number

n balls are thrown into m bins by 2-universal hashing

- Location of n bins: $X_1, X_2, \dots, X_n \in [m]$

Collision #: $Y = \sum_{i < j} I[X_i = X_j]$

- Linearity of expectation:

$$\mathbb{E}[Y] = \sum_{i < j} \Pr[X_i = X_j] \leq \binom{n}{2} \frac{1}{m}$$

2-universal

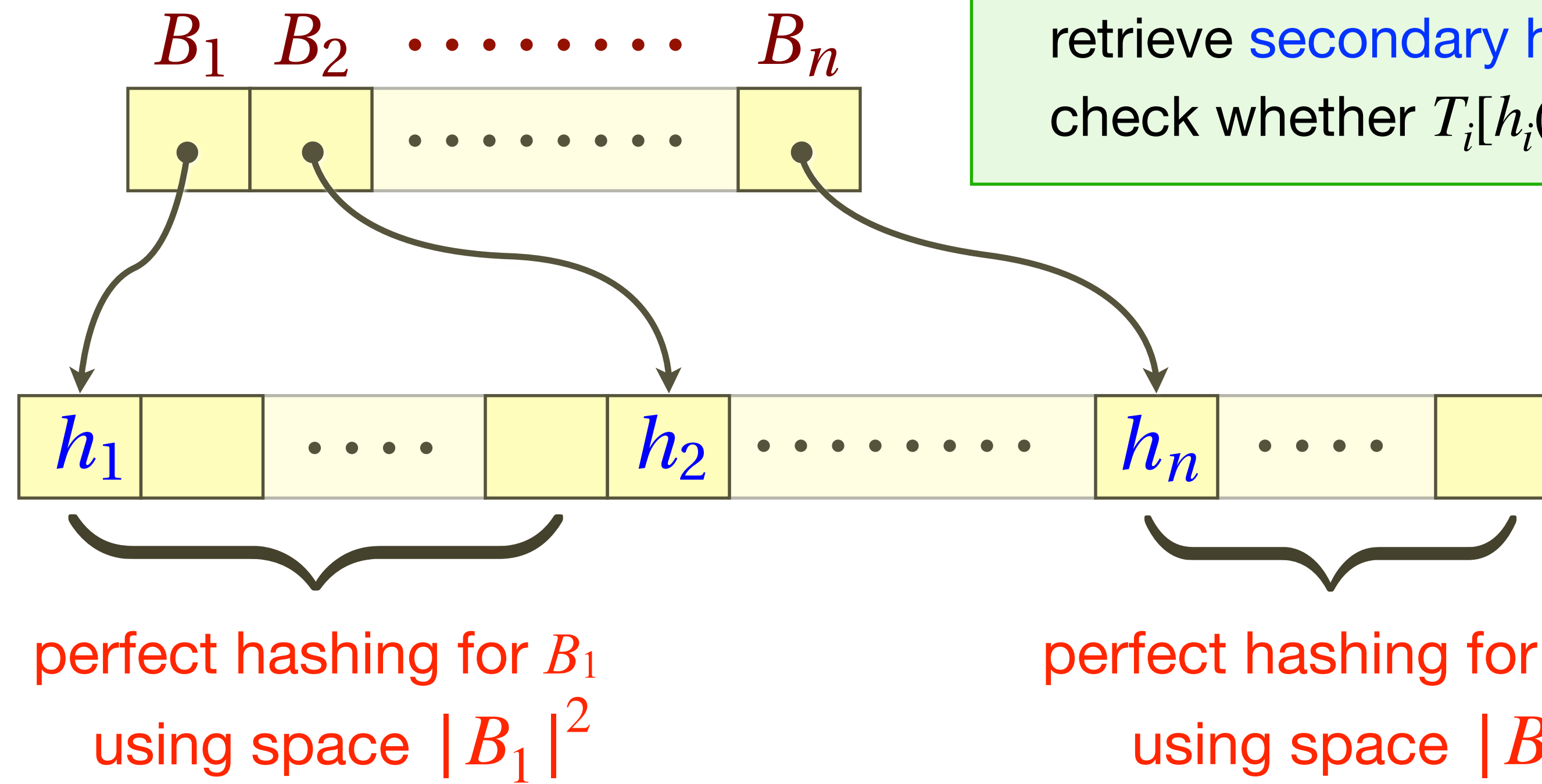
- Size of the i -th bin: $|B_i|$

$$Y = \sum_{i=1}^n \binom{|B_i|}{2} = \frac{1}{2} \sum_{i=1}^n |B_i|(|B_i| - 1) \implies \mathbb{E} \left[\sum_{i=1}^n |B_i|^2 \right] = \frac{n(n-1)}{m} + n$$

FKS Perfect Hashing

Set $S \subseteq [N]$ of size n

$$h: [N] \rightarrow [n]$$



Query(x):

retrieve **primary hash** h ;
goto **bucket** $i = h(x)$;
retrieve **secondary hash** h_i ;
check whether $T_i[h_i(x)] = x$;

- $\exists h$ from a 2-universal family s.t. the total space cost is $O(n)$

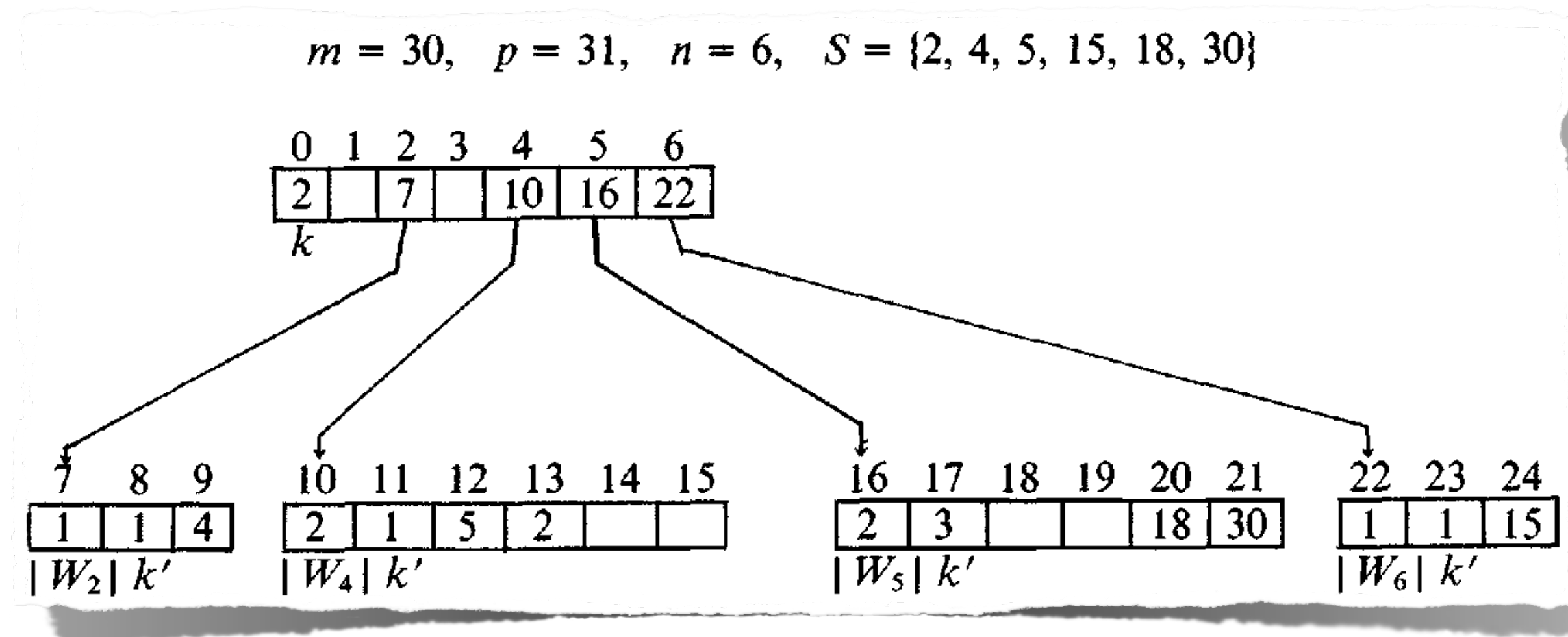
FKS Perfect Hashing

(Fredman, Komlós, Szemerédi, 1984)

Data: a set S of n items $x_1, x_2, \dots, x_n \in U = [N]$

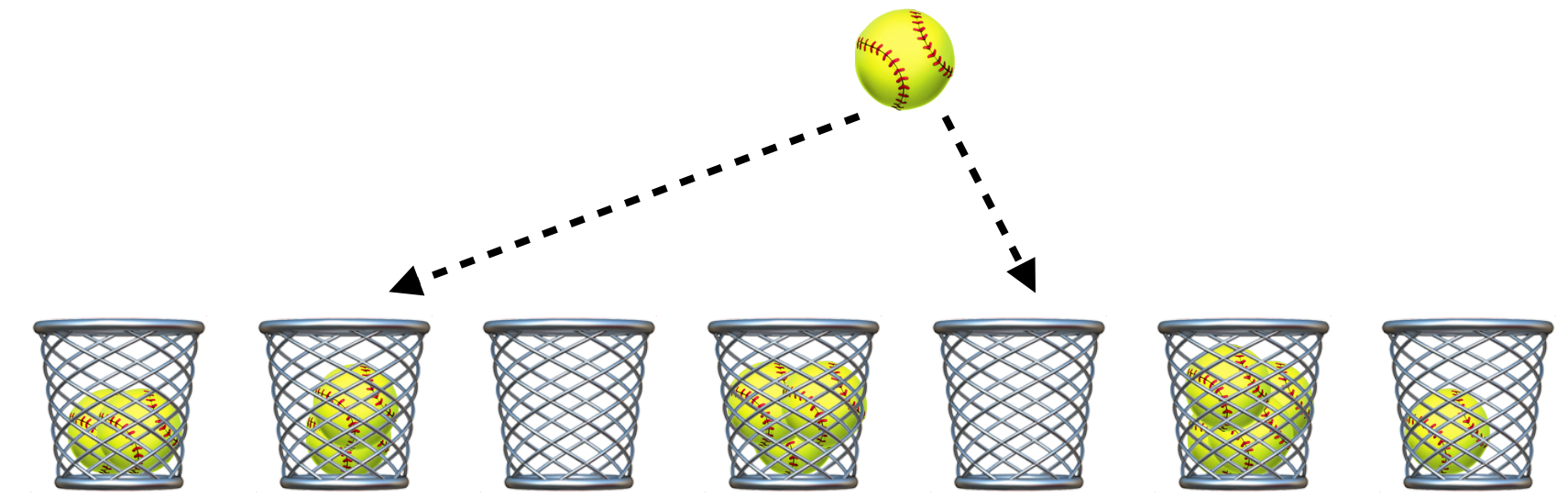
Query: an item $x \in U$

Determine whether $x \in S$.



- $O(n \log N)$ space, $O(1)$ time in the worst case
- Dynamic version: [Dietzfelbinger, Karlin, Mehlhorn, Meyer auf der Heide, Rohnert, Tarjan, 1984]

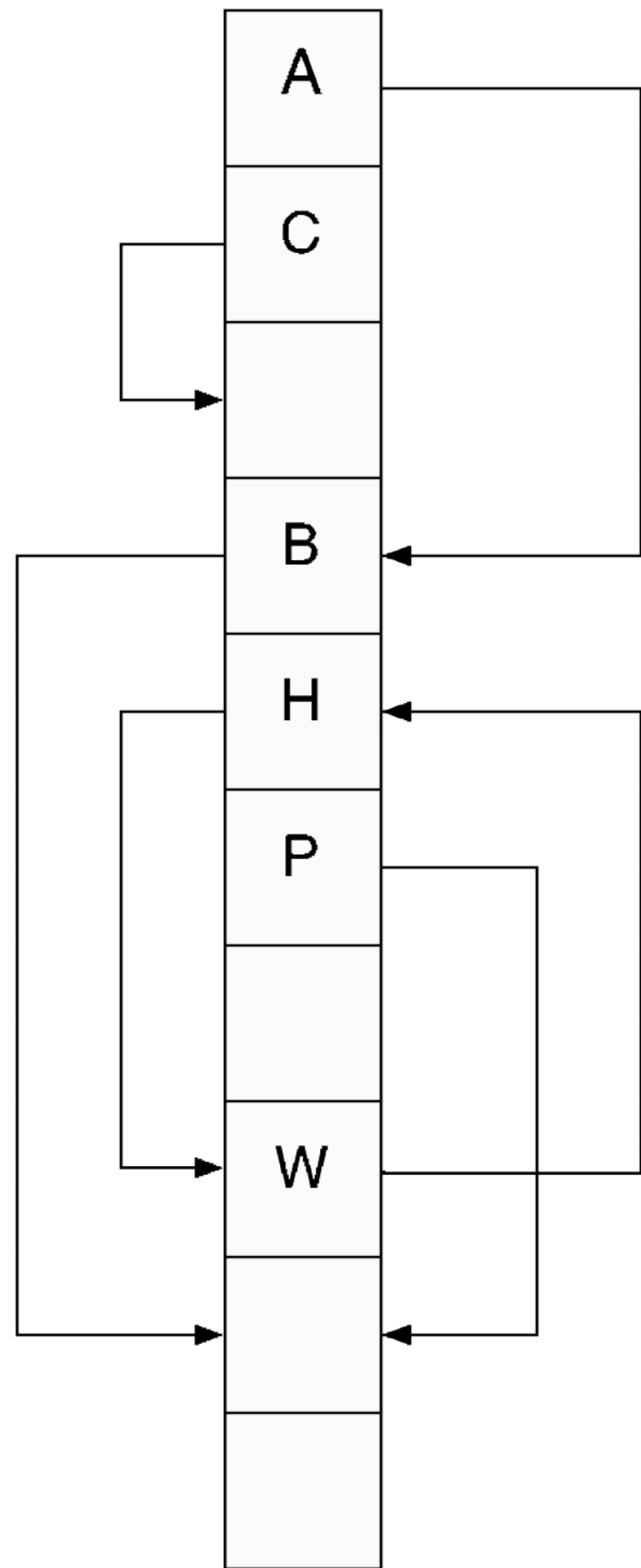
Chaining with two choices



- Maximum load by the power of two choices: $O(\log \log n)$
- $O(\log \log n)$ time for chaining: enumerating because of collisions & max load
- A hash table with two functions?
 - Item x must be placed in $h_1(x)$ or $h_2(x)$, time cost: exact 2
- Collision? Both of two bins are occupied?
 - Kick them away!



Cuckoo Hashing



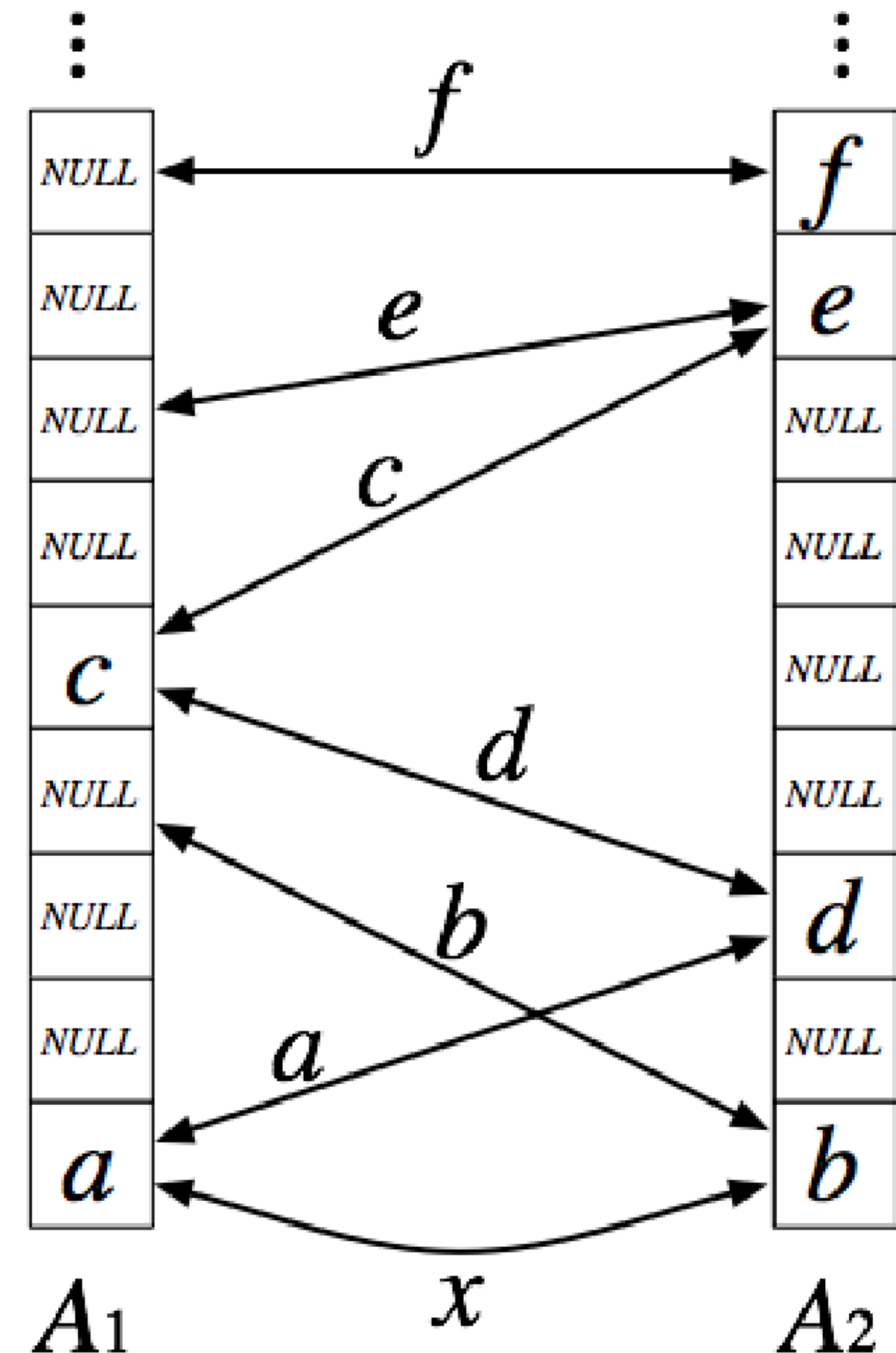
Cuckoo Hashing:

Hash table $A[1 \dots m]$ initialized to empty

Query x : examine $A[h(x)], A[g(x)]$

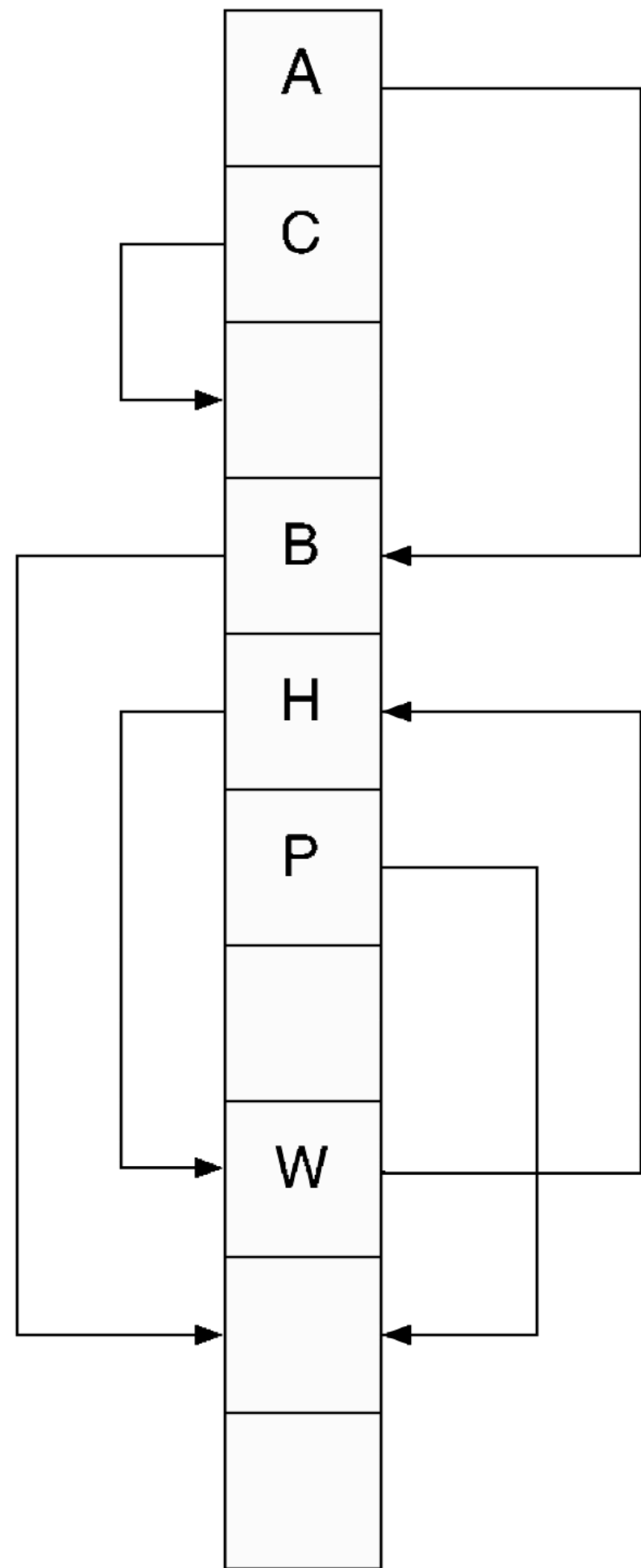
Insert x : place x if exists spare slot;
o.w. while no spare slot, keep kicking;
rebuild if take so long

- Space cost: $O(n)$
- Query time: 2
- Update time: ?



Cuckoo Graph

Analysis of Cuckoo hashing



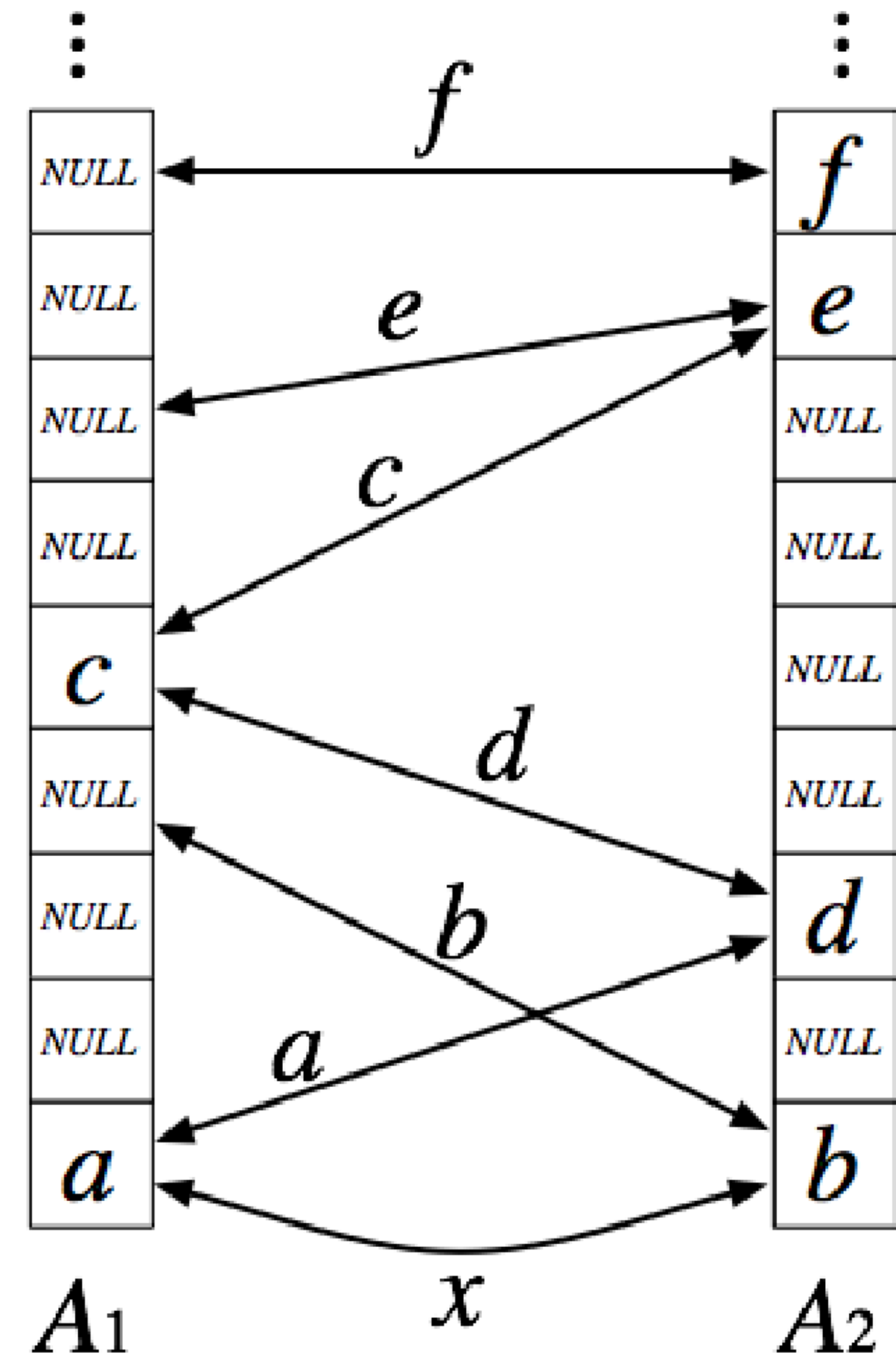
Cuckoo Hashing:

Hash table $A[1 \dots m]$ initialized to empty

Query x : examine $A[h(x)], A[g(x)]$

Insert x : place x if exists spare slot;
o.w. while no spare slot, keep kicking;
rebuild if take so long

- A multigraph $G = (V, E)$
 - V = all the slots
 - $E = \{(h(x), g(x)) : x \in S\}$
 - Edge labeled by insertion



Cuckoo Graph

Analysis of Cuckoo hashing

- A multigraph $G = (V, E)$
 - $V =$ all the slots
 - $E = \{(h(x), g(x)) : x \in S\}$

- When does insertion succeed?

- We have a kicking path:
- Or, a single kicking cycle:

- When does insertion fail?

- A double kicking cycle:
- Long path

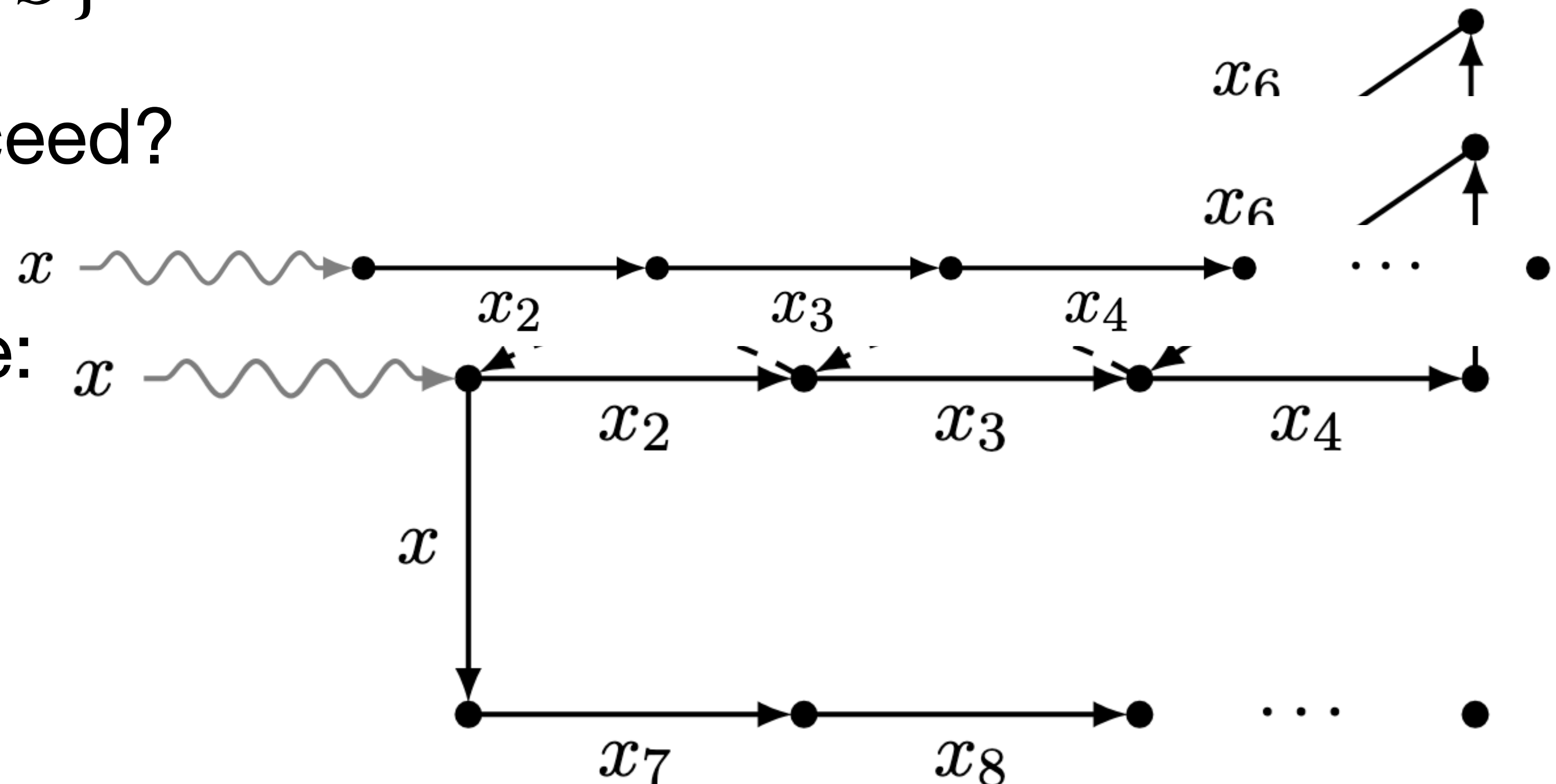
Cuckoo Hashing:

Hash table $A[1..m]$ initialized to empty

Query x : examine $A[h(x)], A[g(x)]$

Insert x : place x if exists spare slot;

o.w. while no spare slot, keep kicking;
rebuild if take so long



Cuckoo Graph

Analysis of Cuckoo hashing

- path 👍
- single cycle 👍
- double cycle 👎

- $\mathbb{E}[T] \leq \mathbb{E}[\text{len}(\text{path})] + \mathbb{E}[\text{len}(\text{cycle})] + \Pr[\text{d-cycle, long path}] \cdot n \cdot \mathbb{E}[T]$
- Claim: A fixed insertion path of length k exists w.p. $\leq m^{-2k-1}$
 - Path be like $x, 1, b, 2, c, 3, e, 4, \dots$ key-slot-key-slot-...
- $\Pr[\text{len}(\text{path}) \geq k] \leq (nm)^k / m^{2k+1} \leq n^k / m^{k+1} = \exp(-\Omega(k))$
- $\Pr[\text{len}(\text{cycle}) \geq k] = \exp(-\Omega(k)), \Pr[\text{path} \geq 2 \log n] \leq 1/n^{\Omega(1)}$
- $\mathbb{E}[\text{len}(\text{path})], \mathbb{E}[\text{len}(\text{cycle})] \leq \sum_{k \geq 1} \exp(-\Omega(k)) = O(1)$

Cuckoo Hashing:

Hash table $A[1 \dots m]$ initialized to empty

Query x : examine $A[h(x)], A[g(x)]$

Insert x : place x if exists spare slot;

o.w. while no spare slot, keep kicking;
rebuild if take so long

Cuckoo Graph

Analysis of Cuckoo hashing

- path 👍
- single cycle 👍
- double cycle 👎
- $\mathbb{E}[T] \leq O(1) + O(1) + (\text{Pr}[\text{d-cycle}] + 1/n^2) \cdot n \cdot \mathbb{E}[T]$
- Claim: A fixed insertion path of length k exists w.p. $\leq m^{-2k-1}$
 - Path be like $x, 1, b, 2, c, 3, e, 4, \dots$ key-slot-key-slot-...
- $\text{Pr}[\text{d-cycle}]?$
 - Observe. Simplest case $\ominus$: $\text{Pr} \leq n^2 m^2 / m^6$.
 - General case $\circ$: $\# \text{edge} \leq \# \text{slot} - 1$. $\text{Pr} \leq n^k m^k / m^{2k+2}$

Cuckoo Hashing:

Hash table $A[1 \dots m]$ initialized to empty

Query x : examine $A[h(x)], A[g(x)]$

Insert x : place x if exists spare slot;

o.w. while no spare slot, keep kicking;
rebuild if take so long

Cuckoo Hashing

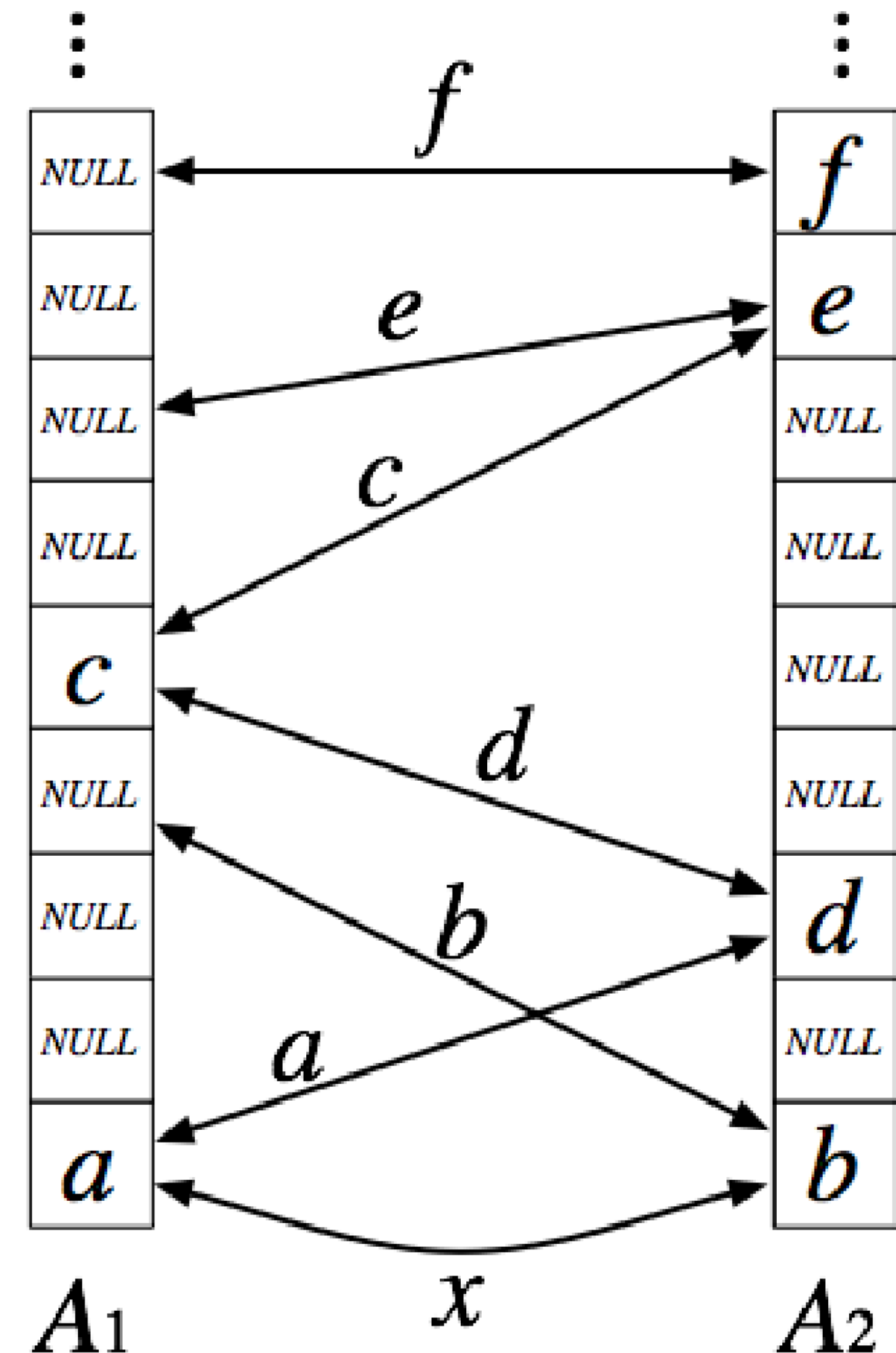
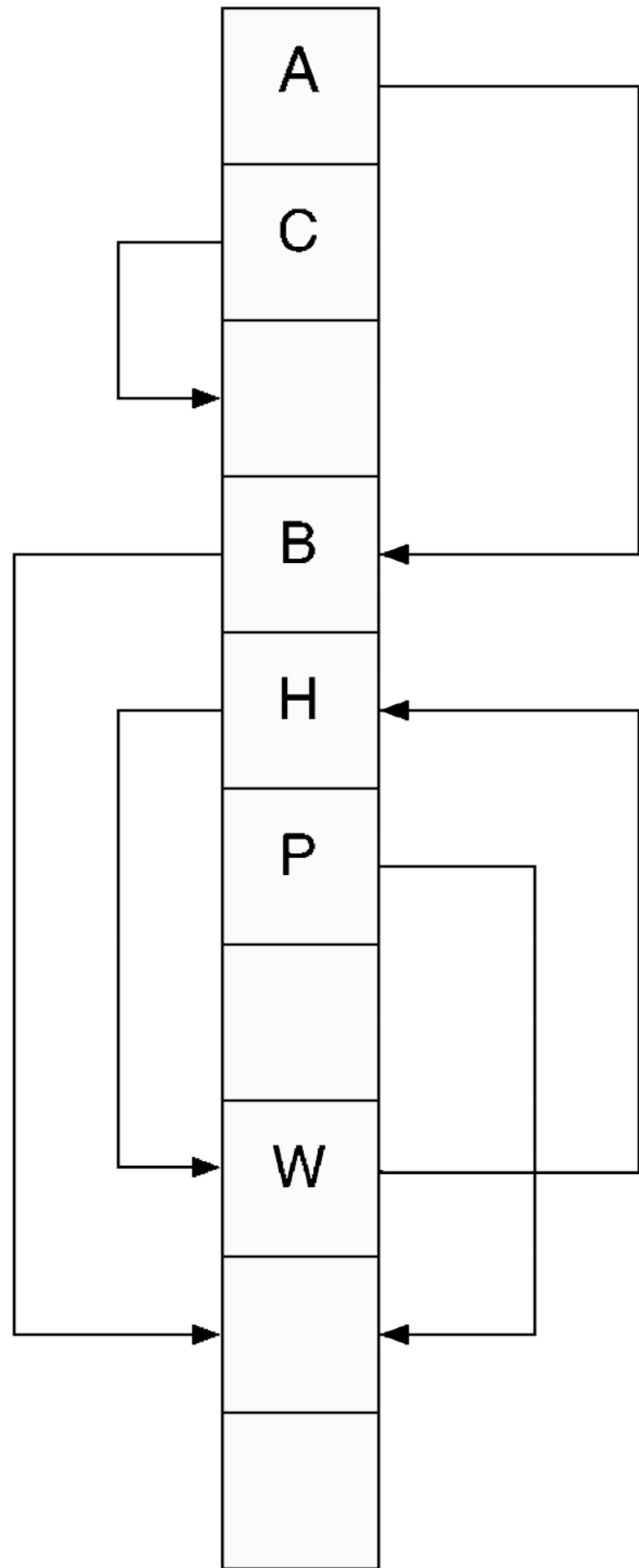
Cuckoo Hashing:

Hash table $A[1 \dots m]$ initialized to empty

Query x : examine $A[h(x)], A[g(x)]$

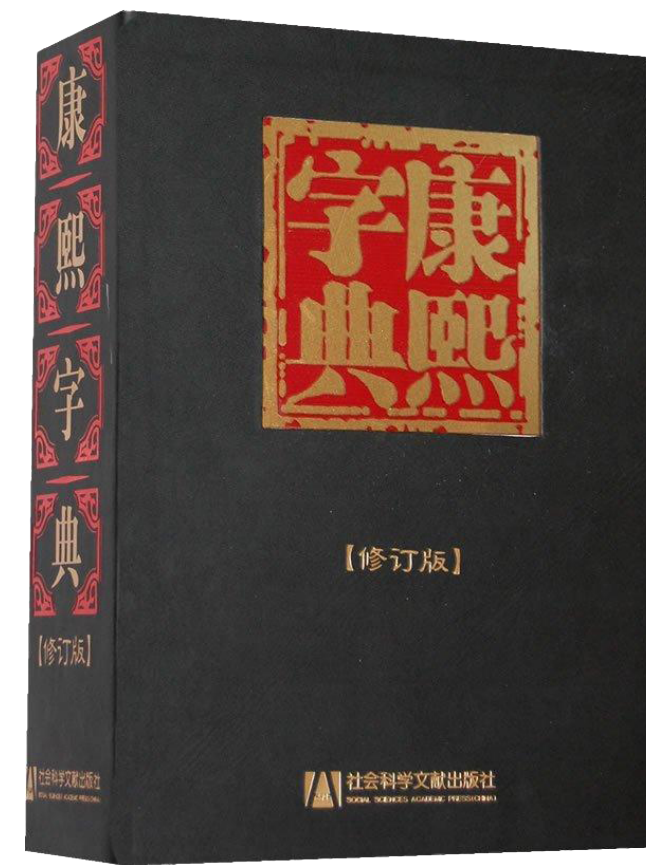
Insert x : place x if exists spare slot;
o.w. while no spare slot, keep kicking;
rebuild if take so long

- Space cost: $O(n)$
- Query time: 2
- Update time: **expected** **amortized** $O(1)$

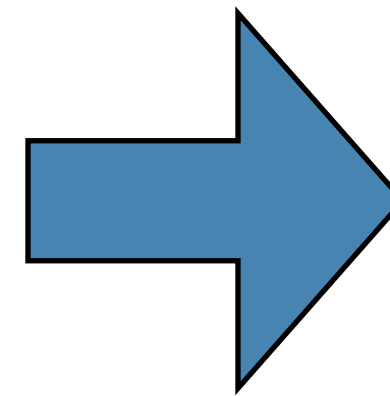


Static Dictionary

Goal: $O(1)$ search time



(word, meaning)-pair



found here

or $x \notin S$

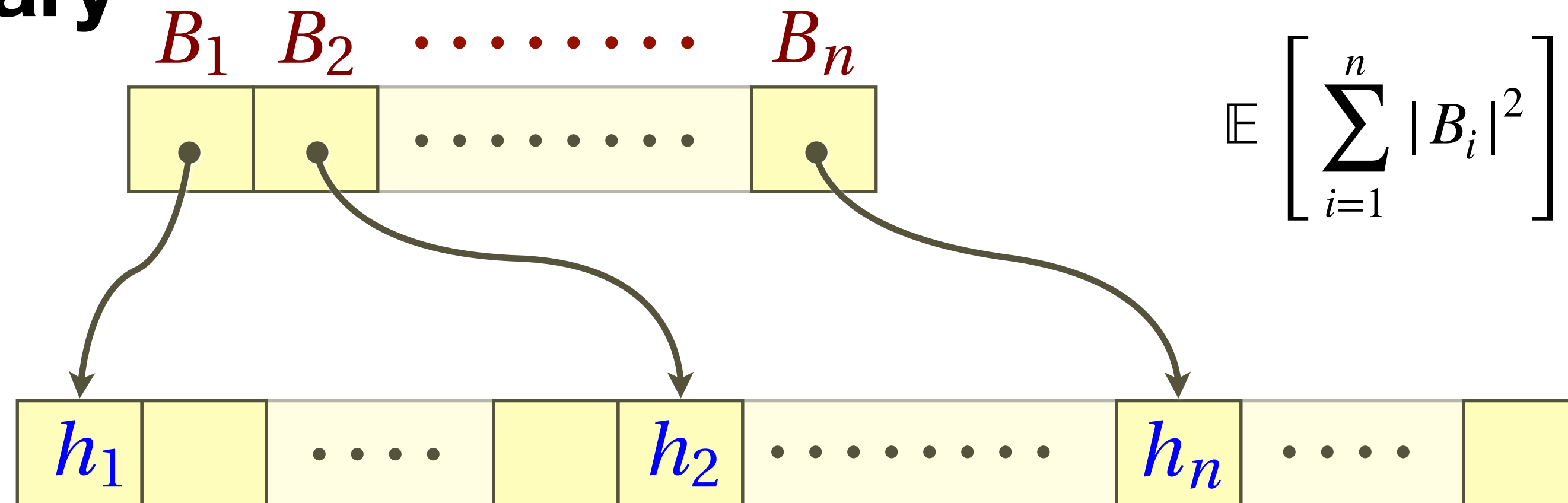
(key, value)-pair

Succinct Dictionaries

	Space	Query time	Update time	
Binary search tree	$O(n)$	$O(\log n)$	$O(\log n)$	bad time
Hashing with Chaining	$O(n)$	expected $O(1)$	expected $O(1)$	bad time
1-level Perfect hashing	$O(n^2)$	1	1	Huge space
2-level Perfect hashing	$O(n)$	$O(1)$	$O(1)$	Huge constants
Cuckoo hashing	$O(n)$	2	expected amortized $O(1)$	worst case?
Succinct Dictionary	$(1+o(1))n$	$O(1)$	$O(1)$	Best of both worlds

Succinct Dictionaries

Small Dictionary



$$\mathbb{E} \left[\sum_{i=1}^n |B_i|^2 \right] = \frac{n(n-1)}{m} + n$$

- Exactly, how much space do we need for FKS hashing?
 - First-level pointers & hashing ($n \log n$ bits) + space for raw data (nv bits)
- Simple observation: it is succinct if $\log n \ll v$
 - Say, $v = \log_2 U$. It is succinct if we use it to maintain $n = \text{poly } \log U$ keys

Succinct Dictionaries

First try

- Examine the balls-into-bins argument:
 - n balls into $n/\log^3 n$ bins. Let $n' \triangleq (1 + 2/\log n)\log^3 n$
 - Chernoff bound: $\Pr[\text{load} \geq n'] \leq \exp(-\mu\delta^2/2) = n^{-2}$
 - Maximum load $\leq (1 + 2/\log n)\log^3 n$ w.h.p.
- Each bin is a small dictionary with redundancy $n' \log n'$ bits
- Overall redundancy $n \log n' + (2/\log n)n \log U = o(n \log U)$ bits

any issue?

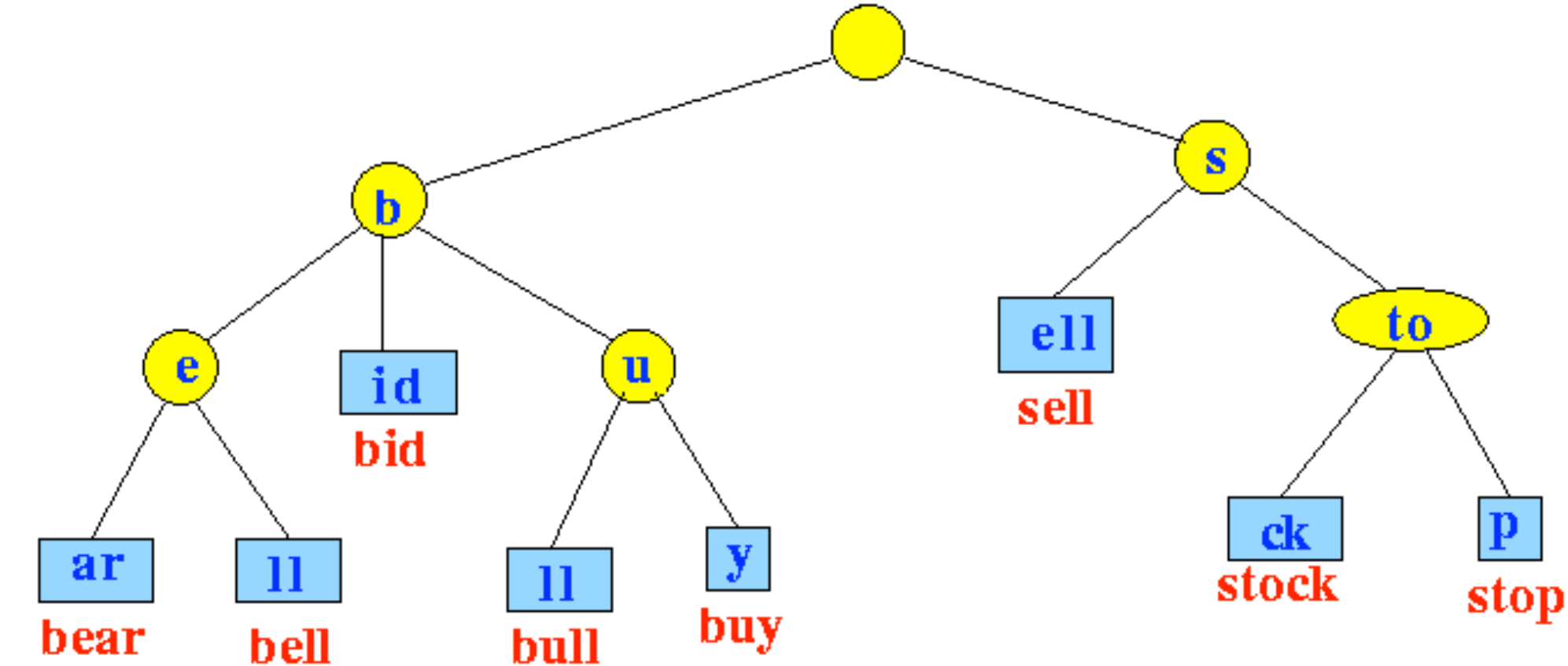
Succinct Dictionaries

Better small dictionary 😊 & Let it overflow!

- Intuitively, we can maintain $O(\log n / \log \log n)$ keys with $O(\log n)$ bits
 - e.g. binary search tree
 - Intuitively, any operation can be done in $O(1)$ times as the word size is $\Omega(\log n)$ bits.
- balls-into-bins argument:
 - n balls into n/m bins, where $m = \log n / \log \log n$
 - Chernoff bound: $\Pr[\text{load} \geq (1 + \delta)m] \leq \exp(-m\delta^2/2) = \log^{-\Omega(1)} n$
 - #overflowing $\approx n / \text{poly } \log n$ *A dictionary maintains $n / \log^9 n$ keys with $n / \log n$ bits*

Succinct Dictionaries

Trie



- We can maintain $O(\log n / \log \log n)$ keys with $O(\log n)$ bits
 - Most basic operations can be done in $O(1)$ times with an $n^{1/3}$ bits precomputed lookup table, since the word size is $\Omega(\log n)$ bits.
- Trie: *Elias-Fano in fact*
 - Compressed trie: upon insertion, reaches leaf, grows until splits
 - Assume random keys
 - One insertion grows k times w.p. 2^{-k} , geometrically.
 - Insertion n' keys, #grows $\sim$ negative binomial.
 - #grows = $\Omega(\log n)$ w.p. $\exp(-\Omega(\log n)) = n^{-\Omega(1)}$

A binary tree of $O(\log n)$ nodes!

encode with $O(\log n)$ bits!

Succinct Dictionaries

Dealing the overflowing

- A dictionary maintains $n/\log^9 n$ keys with $n/\log n$ bits
 - n' balls into n' bins, where $n' = n/\log^9 n$.
 - $\Pr[\text{load} \geq m] \leq \sum_{k \geq m} \binom{n'}{k} \left(\frac{1}{n'}\right)^k = \exp(-\Omega(m \log n'))$
 - Let $m = \log n / \log \log n$, the probability is $\exp(-\Omega(\log n))$
 - Apply the trie
- Overall space: $O(n' \log n) = O(n/\log n)$ bits

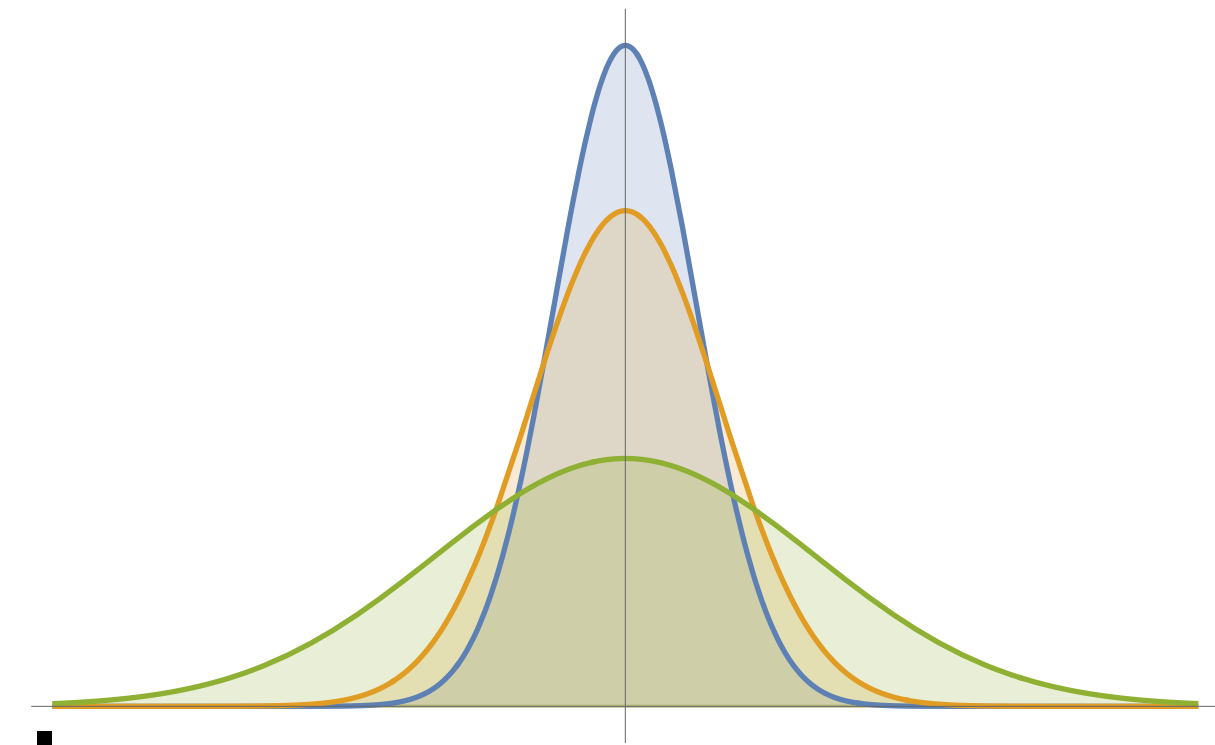
Succinct Dictionaries

Better small dictionary 😊 & Let it overflow!

- We can maintain $O(\log n / \log \log n)$ keys with $O(\log n)$ bits
 - Most basic operations can be done in $O(1)$ times with an $n^{1/3}$ bits precomputed lookup table, since the word size is $\Omega(\log n)$ bits.
- balls-into-bins argument:
 - n balls into n/m bins, where $m = \log n / \log \log n$
 - Chernoff bound: $\Pr[\text{load} \geq (1 + \delta)m] \leq \exp(-m\delta^2/2)$ Let $\delta = \Omega(\log \log n / \sqrt{\log n})$
 - #overflowing $\approx n / \text{poly } \log n$ A dictionary maintains $n / \log^9 n$ keys with $n / \log n$ bits
 - Redundancy: $O(\delta n)$ words $O(\delta n \log(u/n))$ bits by Feistel

Hashing in Practice

Chernoff Bound



- Chernoff bound: Let $X_1, \dots, X_n \in \{0,1\}$ be **independent** trials

$$X = \sum_{i=1}^n X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

(Poisson binomial RV
with mean μ)

- For any $\delta > 0$,

$$\Pr(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$$

- For any $0 < \delta < 1$,

$$\Pr(X \leq (1 - \delta)\mu) \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu$$

Chernoff Bound With Limited Independency

Save randomness, save money

- Chernoff bound: Let $X_1, \dots, X_n \in \{0,1\}$ be **k -wise** independent trials

$$X = \sum_{i=1}^n X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

- For any $\delta > 0$, $k \leq \frac{\mu\delta}{1 - \mu/n}$,

$$\Pr(X \geq (1 + \delta)\mu) \leq \frac{\binom{n}{k} (\mu/n)^k}{\binom{(1 + \delta)\mu}{k}} = \left(\frac{\mu}{n}\right)^k \cdot \prod_{i=0}^{k-1} \frac{n - i}{(1 + \delta)\mu - i}$$

A closer look at Chernoff

- Why do we like moment generating function?

- $M_x(t) = \mathbb{E} [e^{tX}] = \sum_{i \geq 0} t^i \mathbb{E} [X^i] / i!$ exhausts all moments

- What is k -th moment of sum of Bernoulli R.V.?

- $X^1 = \sum_i X_i, X^2 = \sum_i X_i^2 + \sum_{i \neq j} X_i X_j = X^1 + \sum_{i \neq j} X_i X_j$

- $X^3 = \sum_i X_i^3 + \sum_{i \neq j} (X_i X_j^2 + X_i^2 X_j) + \sum_{i \neq j \neq k} X_i X_j X_k = \sum_i X_i + 2 \sum_{i \neq j} X_i X_j + \sum_{i \neq j \neq k} X_i X_j X_k$

- Intuition?

- $X^k = \sum_{i \leq \min\{k, n\}} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} X_{j_1} X_{j_2} \dots X_{j_i}$

A closer look at Chernoff

- Why do we like moment generating function?

- $M_x(t) = \mathbb{E} [e^{tX}] = \sum_{i \geq 0} t^i \mathbb{E} [X^i] / i!$ exhausts all moments

- Intuition?

- $e^{tX} = \sum_{i \leq \min\{k, n\}} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} X_{j_1} X_{j_2} \dots X_{j_i}$ for some $\alpha_1, \dots, \alpha_n \geq 0$

- $\Pr[X \geq a] = \Pr \left[\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} X_{j_1} X_{j_2} \dots X_{j_i} \geq \sum_{i=0}^a \alpha_i \binom{a}{i} \right]$

Generalized Chernoff

- $\Pr[X \geq a] = \Pr \left[\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} X_{j_1} X_{j_2} \dots X_{j_i} \geq \sum_{i=0}^a \alpha_i \binom{a}{i} \right]$
- For Chernoff, the α_i 's are fixed s.t. $\sum_{i \leq n} \alpha_i \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} X_{j_1} X_{j_2} \dots X_{j_i} = \sum_{i \leq n} \frac{t^i X^i}{i!}$
- Can we do better from this perspective?
 - Parameterize α_i 's instead of simply t

Generalized Chernoff

- $\Pr[X \geq a] = \Pr \left[\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} X_{j_1} X_{j_2} \dots X_{j_i} \geq \sum_{i=0}^a \alpha_i \binom{a}{i} \right]$
 - Parameterize α_i 's instead of simply t
- Markov first: $\Pr[X \geq a] \leq \frac{\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_i}]}{\sum_{i=0}^a \alpha_i \binom{a}{i}}$
- Observation: for $i > a$, α_i should be 0
- Observation: α_i should be 0 for i not minimizes $\sum \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_i}] / \binom{a}{i}$

Generalized Chernoff

Parameterize α_i 's instead of simply t

- $\Pr[X \geq a] \leq \frac{\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_i}]}{\sum_{i=0}^a \alpha_i \binom{a}{i}}$
- Observation: $\sum \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_i}]$ maximized when $p_1 = \dots = p_n$
subject to $\sum_i p_i = \mu$
 - $\sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_i}] \leq \binom{n}{i} (\mu/n)^i$ if mutual independence

Generalized Chernoff

Parameterize α_i 's instead of simply t (assume independence)

- $\Pr[X \geq a] \leq \frac{\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_i}]}{\sum_{i=0}^a \alpha_i \binom{a}{i}} \leq \frac{\sum_{i=0}^a \alpha_i \binom{n}{i} (\mu/n)^i}{\sum_{i=0}^a \alpha_i \binom{a}{i}}$

- Observation: α_i should be 0 for i not minimizes $\binom{n}{i} (\mu/n)^i / \binom{a}{i}$

- Observation: $\frac{\binom{n}{i+1} (\mu/n)^{i+1}}{\binom{a}{i+1}} / \frac{\binom{n}{i} (\mu/n)^i}{\binom{a}{i}} = \frac{n-i}{a-i} \frac{\mu}{n}$, minimizes when
 $i \triangleq (a - \mu)/(1 - \mu/n)$

Generalized Chernoff

Parameterize α_i 's instead of simply t (assume independence)

- $\Pr[X \geq a] \leq \frac{\sum_{i=0}^a \alpha_i \binom{n}{i} (\mu/n)^i}{\sum_{i=0}^a \alpha_i \binom{a}{i}} \leq \binom{n}{k} (\mu/n)^k / \binom{a}{k}$ for all $k \leq i^*$
- Observation: α_i should be 0 for i not minimizes $\binom{n}{i} (\mu/n)^i / \binom{a}{i}$
- Observation: minimizes when $i^* \triangleq (a - \mu)/(1 - \mu/n)$

Generalized Chernoff

Parameterize α_i 's instead of simply t (k -wise independence)

- $\Pr[X \geq a] \leq \frac{\sum_{i=0}^a \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq n} \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_i}]}{\sum_{i=0}^a \alpha_i \binom{a}{i}} \leq \frac{\sum_{i=0}^a \alpha_i \binom{n}{i} (\mu/n)^i}{\sum_{i=0}^a \alpha_i \binom{a}{i}}$
 $\leq \binom{n}{k} (\mu/n)^k / \binom{a}{k}$ for all $k \leq i^*$
- Observation: α_i should be 0 for $i \neq i^* \triangleq (a - \mu)/(1 - \mu/n)$
- Observation: $\sum \mathbb{E} [X_{j_1} X_{j_2} \dots X_{j_k}] \leq \binom{n}{k} (\mu/n)^k$ with k -wise independence

Chernoff Bound With Limited Independency

Save randomness, save money

- Chernoff bound: Let $X_1, \dots, X_n \in \{0,1\}$ be **k -wise** independent trials

$$X = \sum_{i=1}^n X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

- For any $\delta > 0$, $k \leq \frac{\mu\delta}{1 - \mu/n}$,

$$\Pr(X \geq (1 + \delta)\mu) \leq \frac{\binom{n}{k} (\mu/n)^k}{\binom{(1 + \delta)\mu}{k}} = \left(\frac{\mu}{n}\right)^k \cdot \prod_{i=0}^{k-1} \frac{n - i}{(1 + \delta)\mu - i}$$

Is universal hashing a good notion?

At what cost we leverage independence?

- Time cost for evaluating k -universal hashing: $O(k)$
- Space cost: $O(k)$
- How much independence do we need?
 - Chaining / Maximum load: $\Theta(\log n / \log \log n)$
 - Linear probing: ≤ 5 [PR'07], ≥ 5 [PT'10]
 - Cuckoo hashing: $O(\log n)$
 - F_2 estimation: 4 [AMS'99]
 - min-wise independence (Jaccard similarity $J(A, B) = |A \cap B| / |A \cup B|$): $O(\log(1/\epsilon))$ [Indyk'00], $\Omega(\log(1/\epsilon))$ [PT'10]

Is universal hashing a good notion?

At what cost we leverage independence?

- Time cost for evaluating k -universal hashing: $O(k)$
 - Space cost: $O(k)$
 - Can we do better?
 - Not really [Siegel'89]:
 - $t < k$ time requires $|U|^{1/t}$ space;
 - $\Omega(1)$ -universality in $O(1)$ time requires $|U|^{\Omega(1)}$ space
 - Why?
 - Slow evaluating both in theory and in practice
- But we care about only practice
- A fast hashing in practice?

Tabulation hashing [Zobrist'79]

Bitwise operation is fast

- A hashing function based on bitwise operation?
- Decompose $x = (x_1, \dots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \dots, T_c : [u] \rightarrow [m]$
- Hash value:

$$h(x) = T_1[x_1] \oplus \dots \oplus T_c[x_c]$$

- Space cu , time $O(c)$
- Fits in cache for $c = 4$, $u = 256$, $u^c = 2^{32}$. Extremely fast in practice.

Tabulation hashing

Bitwise operation is fast

- Decompose $x = (x_1, \dots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \dots, T_c : [u] \rightarrow [m]$
- Hash value:

$$h(x) = T_1[x_1] \oplus \dots \oplus T_c[x_c]$$

- Space cu , time $O(c)$
- Tabulation hashing is 3-universal
- But is not even 4-universal: $h(a_1, a_2) \oplus h(a_1, b_2) \oplus h(b_1, a_2) \oplus h(b_1, b_2) = 0$

Tabulation hashing

Bitwise operation is fast

- Decompose $x = (x_1, \dots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \dots, T_c : [u] \rightarrow [m]$
- Hash value: $h(x) = T_1[x_1] \oplus \dots \oplus T_c[x_c]$ is not even 4-universal
- **But** works for all the aforementioned applications:
 - Chaining / Maximum load: $\Theta(\log n / \log \log n)$
 - Linear probing: ≤ 5 [PR'07], ≥ 5 [PT'10]
 - Cuckoo hashing: $O(\log n)$
 - F_2 estimation: 4 [AMS'99]
 - min-wise independence: $O(\log(1/\epsilon))$ [Indyk'00], $\Omega(\log(1/\epsilon))$ [PT'10]

Tabulation hashing

- Decompose $x = (x_1, \dots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \dots, T_c : [u] \rightarrow [m]$
- Hash value: $h(x) = T_1[x_1] \oplus \dots \oplus T_c[x_c]$ is not even 4-universal
- Most probabilistic analysis is based on independence
 - When is $h(x)$ independent of $h(y_1), h(y_2), \dots$?
 - Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$
 - A set S is mutually independent if $\forall x \in S, \exists$ good position character

Peelability

First try

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S, \exists$ good position character
- **Claim:** For any $|S| = d$, $\exists$ peelable subset of size $t \geq \max\{d^{1/c}, \log_2 d\}$
- $n \leq m^{1-\epsilon}$ balls into m bins
 - Any set $|S| \geq d$ lands into a common bin w.p. $1/m^{t-1}$
 - $\binom{n}{t} < n^t$ possible sets. Max load by union bound: w.p. $n^t/m^{t-1} \leq m^{1-\epsilon t}$

Peelability

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S, \exists$ good position character
- **Claim:** For any $|S| = d$, $\exists$ peelable subset of size $t \geq \max\{d^{1/c}, \log_2 d\}$
- $t \geq d^{1/c}$
 - $S \subseteq S_1 \times S_2 \times \dots \times S_c$, there is $i \in [c]$ s.t. $|S_i| \geq |S|^{1/c}$
- $t \geq \log_2 d$
 - Recursion: as long as $|S| > 1$, we can pick from $|S_i| > 1$

Peeling

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S, \exists$ good position character
- Idea:
 - Find a good PC α , then S_α is independent of $S_{-\alpha}$
 - Keep finding, get independent sets $S_\alpha, S_\beta, \dots$
 - Apply peelability argument on S_α, S_β respectively
 - Improvement overall

Anything wrong?

Peeling

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S, \exists$ good position character
- Idea:
 - Find a good PC α , then S_α is independent of $S_{-\alpha}$
 - Keep finding, get **non**-independent sets $S_\alpha, S_\beta, \dots$
 - $\mathbb{E}[X_\alpha | X_\beta = x_\beta, \dots] = \mathbb{E}[X_\alpha]$

A closer look at Chernoff (assume $x_i \in [0,1]$)

- We are examining in fact

(choose $t = \ln(1 + \delta)$)

$$\Pr[X \geq (1 + \delta)\mu] \leq \Pr[e^{tX} \geq e^{t(1+\delta)\mu}] = \Pr[(1 + \delta)^X \geq (1 + \delta)^{(1+\delta)\mu}]$$

- $\mathbb{E}[(1 + \delta)^X] = \sum_{x_{<n}} \left(\Pr[X_{<n} = x_{<n}] \cdot \mathbb{E}[(1 + \delta)^X \mid X_{<n} = x_{<n}] \right)$
 $= \sum_{x_{<n}} \left(\Pr[X_{<n} = x_{<n}] \cdot (1 + \delta)^{\sum_{i<n} x_i} \cdot \mathbb{E}[(1 + \delta)^{X_n} \mid X_{<n} = x_{<n}] \right)$
- $\mathbb{E}[(1 + \delta)^Y] \leq \mathbb{E}[Y] \cdot (1 + \delta)^1 + (1 - \mathbb{E}[Y]) \cdot (1 + \delta)^0 = 1 + \delta \mathbb{E}[Y] \leq e^{\delta \mathbb{E}[Y]}$
- $\mathbb{E}[(1 + \delta)^X] \leq \sum_{x_{<n}} \left(\Pr[X_{<n} = x_{<n}] \cdot (1 + \delta)^{\sum_{i<n} x_i} \cdot e^{\delta \mathbb{E}[X_n]} \right) \leq e^{\delta \mathbb{E}[X]}$

Peeling

Chernoff: For $X_1, \dots \in [0, d]$ with fixed means,

$$\Pr \left[\sum X_i \geq (1 + \delta)\mu \right] \leq \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^{\mu/d}$$

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S, \exists$ good position character
- Idea:
 - Find a good PC α , then S_α is independent of $S_{-\alpha}$
 - Keep finding, get **non**-independent sets $S_\alpha, S_\beta, \dots$
 - $\mathbb{E}[X_\alpha | X_\beta = x_\beta, \dots] = \mathbb{E}[X_\alpha]$
 - Upper bound? **maximum load!**

Peeling

Chernoff: For $X_1, \dots \in [0, d]$ with fixed means,

$$\Pr \left[\sum X_i \geq (1 + \delta)\mu \right] \leq \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^{\mu/d}$$

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S, \exists$ good position character
 - Find a good PC α , then S_α is independent of $S_{-\alpha}$ **and** $|S_\alpha| \leq n^{1-1/c}$
 - Keep finding, get **non**-independent sets $S_\alpha, S_\beta, \dots$ **with fixed means**
 - Max-load: for any bin, d balls from S_α land w.p. $m^{1-t/c}$
 - w.p. $1 - n \cdot m^{1-t/c}$, $X_\alpha \leq d$ for all α

max-load: $n \leq m^{1-\epsilon}$ balls into m bins,
max-load d w.p. $\leq m^{1-\epsilon t}$